

ORIGINAL ARTICLE

The Law is the Last to Know: Evidence That De Facto Progress Drives De Jure Women's Rights

Joshua D. Ammons¹  | Daniel J. D'Amico²

¹The Stephenson Institute for Classical Liberalism, Wabash College, Crawfordsville, Indiana, USA | ²The Stephenson Institute and Affiliated Associate Professor, Wabash College, Crawfordsville, Indiana, USA

Correspondence: Joshua D. Ammons (ammonsj@wabash.edu)

Received: 28 August 2025 | **Revised:** 15 December 2025 | **Accepted:** 5 January 2026

Keywords: cultural norms | economic development | economic freedom | gender equality | institutional change | legal rights | property rights

ABSTRACT

This paper examines how improvements in women's formal legal rights shape economic and social outcomes across countries. Using the Gender Disparity Index from the Economic Freedom of the World dataset, we analyze cases where countries experienced substantial increases in women's legal rights sustained for at least 5 years. Contrary to expectations, our findings show that such policy changes did not yield better economic and social outcomes. To explain these results, we demonstrate that the average annual gains in de facto property rights for women were greatest before these changes were codified into law. Formal legal advances follow, rather than drive, informal changes in women's social participation and economic roles. This study emphasizes the informal dependency of institutional change and its implications for economic development and social welfare.

“The loss to the world, by refusing to make use of one-half of the whole quantity of talent it possesses, is extremely serious. The result is an economy of force which is not only barren but suicidal, throwing away on the mere command over unwilling slaves the energies which, if exerted in any other direction, would produce an indefinite increase of human good.”

- John Stuart Mill (1879).

“Don't sit down and wait for the opportunities to come. Get up and make them.”

- Madam C. J. Walker (1912) ¹

de facto behavioral patterns and social processes that individuals and groups previously secured via extra-legal channels? This question is particularly important when examining women's rights, as significant variations exist between de jure legal rights and de facto economic and social opportunities for women across countries and time periods. Given the established theory and evidence regarding property rights and inclusive institutions on the one hand; and the outcomes of development, economic growth, and human flourishing on the other (Gwartney 2009); like other markers of economic openness and inclusive institutions (Acemoğlu and Robinson 2016), we expect to observe beneficial effects after a substantial enhancement of economic rights for women. Akin to Mill's masthead quotation, we further expect such effects to be large, as gender-exclusionary policies apply to roughly half of any population within a respective country.

2 | Introduction

Do de jure policy changes and legal reforms drive economic and social progress? Or instead, do legislative changes merely reflect

Our study examines how significant improvements in women's formal legal rights are correlated with various economic and social outcomes. Using the Gender Disparity Index from the Economic Freedom of the World dataset, which measures legal equality between men and women across countries,

we analyze cases where countries experienced substantial increases in women's legal rights that persisted for at least 5 years. We then examine how these changes shaped outcomes relevant to female and overall social well-being, including GDP per capita, maternal mortality, urban sanitation, and life expectancy.

The core economic rights measured in the Gender Disparity Index encompass several fundamental areas of formal economic and social participation for women. These include a woman's right to own and inherit property on equal terms with men, her ability to work in any profession or trade without requiring permission, her freedom to conduct financial transactions such as opening bank accounts and signing contracts independently, her capacity to register and operate businesses, her freedom to travel both domestically and internationally, and her right to make independent legal decisions, including initiating legal proceedings and serving as the head of a household. We accept these measures as viable proxies for the basket of rights broadly representing the essential freedoms necessary for women to participate independently in formal economic activities.²

Contrary to our expectations and conventional understandings of the positive relationship between inclusive formal institutions and economic performance, we find no large positive effects from enhanced property rights for women, nor can we establish any systematic, or even small, positive effect from enhanced formal property rights for women. Major improvements in women's formal legal rights are not correlated with better economic or social outcomes. In fact, with preliminary methods, some of our measures of social well-being showed negative outcomes. These results demand explanation.

Drawing from a variety of institutional theory and evidence, we argue that de jure legal changes may represent the culmination of, rather than the catalyst for, women's material and social advancement. Rational actors discover and innovate methods to pursue and accomplish their plans, even in the face of restrictive policies and prohibitions. Hence, codifying rights to forms of social behavior and economic participation are likely to carry only marginal, non-existent, or even negative influences upon generalized economic and social performance. Presumably, economic and social benefits already accrued through rational but informal past efforts when agents in the society strategized rationally within the non-gender-inclusive legal institutions.

In so far as past agents were likely to seek the greatest returns for their restricted behaviors and institutional investments, we expect the resultant measures of economic and social progress to conform to the law of diminishing returns. Second, in so far as female exclusion was a long-standing institutional practice prior to formal inclusive reforms; the effects on economic and social welfare from enhancing legal inclusion for women in late historical periods would tend to be marginal. Hence, when we observe contemporary reforms toward inclusivity within environments with long-standing gender exclusion, much of the economic and social returns of de facto female participation have already manifested in the past, and thus do not appear significant in our data. We empirically examine this hypothesis by comparing results using V-dem's measure of property rights

for women, which encompasses de facto and de jure rights, with the measure of de jure rights in the Economic Freedom of the World Index.

To investigate the impacts of de jure improvements in female property rights, we employ matching methods and Callaway and Sant'Anna (2021). We supplement these approaches with detailed examinations of how our results vary with different thresholds for what constitutes a significant legal change. By examining these relationships through multiple methodological lenses, we aim to provide robust evidence about how de jure legal rights may interact with individual and collective actions across formal and informal institutional settings.

Our findings contribute to ongoing debates about institutional change and economic development by highlighting how informal processes often undergird and may serve as a foundation for formal legal reforms to advance women's social status. This pattern is further evidenced in organizational contexts, where even in industries dominated by women, gender dynamics continue to shape leadership and risk-taking behaviors (Rinne and Sonnabend 2021). Our results have important implications for policymakers and advocates working to promote gender equality. First, effective institutional change is a complex and uncertain process, likely subject to similar limitations as central planning in general. Unintended consequences may offset expected returns. In short, the precision and accuracy needed to target women as a subgroup population for economic and social advancement may not be complemented by the limited potential of formal reform efforts. Instead, reform efforts that bolster women's de facto economic and social participation rates and informal cultural networks may be more consequential than de jure legal changes. Furthermore, and in line with established evidence (Kramer 2024; Berggren and Bjørnskov 2024), converging upon institutional types and reforms that assure generalized economic performance, regardless of gender, may better promote the material well-being and social status of women as a sub-group within a country over legal policies that specifically aim at progressing women's legal rights.

This paper proceeds in seven main sections. Section 2 provides a comprehensive review of relevant literature on economic freedom and the gender disparity index. Section 3 outlines our theory. Section 4 describes the dataset used in the analysis, including its sources, characteristics, and any pre-processing steps taken. Section 5 outlines the methodological approach employed to address the research questions, detailing the statistical techniques and models used. In Section 6, we outline our results. Finally, Section 7 summarizes the key findings, discusses their implications, and suggests possible explanations.

3 | Literature Review

The Economic Freedom of the World (EFW) index, as described by Gwartney et al. (2021), measures the degree to which policies and institutions in a country support economic freedom. This index has been widely used to investigate the effects of economic freedom upon a variety of relevant economic and social outcomes, including investment, economic growth, income levels, and poverty rates. Gwartney et al. (2021) report that nations

with higher levels of economic freedom consistently demonstrate higher investment rates, faster economic growth, higher income levels, and faster reductions of poverty.

The relationship between economic freedom and gender equality has recently received increased scholarly attention. Kramer (2024) provides compelling evidence that countries with higher levels of economic freedom exhibit fewer legal differences between men's and women's economic rights, particularly related to work opportunities, compensation rates, and levels of entrepreneurship. This research challenges the notion that market-based systems disadvantage women, instead suggesting that economic freedom tends to at least correlate with more equitable legal treatment across genders. In support of this perspective, Davis and Williamson (2019) show that societies that emphasize individual rights and autonomy tend to demonstrate greater gender equality on multiple dimensions, including employment, income, education, and leadership positions.

In business regulation, Bastos and Pavlik (2024) investigate whether firms with female ownership experience different regulatory burdens than their male counterparts. Their findings indicate that female-owned firms report higher time costs associated with regulatory compliance and face longer waiting periods for construction permits. These disparities are most pronounced in countries with the most significant gender disparity in *de jure* rules, highlighting the importance of both legal frameworks and their practical implementations.

Addressing concerns about capitalism's impact on women, Berggren and Bjørnskov (2024) employ an empirical approach to examine the relationship between economic freedom and life satisfaction across genders. Their research finds that most forms of economic freedom, particularly legal quality, monetary stability, openness, and regulation, benefit general life satisfaction. Importantly, they observe no significant differences between men and women in these benefits, challenging the notion that capitalism inherently disfavors women in terms of life satisfaction.

In the developing world context, Nziku and Henry (2021) critically examine women's entrepreneurship policies in Tanzania. Their study reveals that while current policies acknowledge women's important role in communities, they are often geared more toward small and medium-sized enterprise (SME) development than entrepreneurship. The authors argue for more contextualized and gender-sensitive policies to effectively support women entrepreneurs in specific geographical and cultural settings.

4 | Theory

The evolution of women's legal rights presents an illuminating case study in institutional change. Akin to Mill (1879)'s insights above and our initial intuitions, one interpretation of modern institutional theory implies that formal legislation that secures women's economic rights would thus instigate and or substantially improve subsequent rates of female economic and social participation. With such heightened participation one would

expect also measurable increases in economic and social well-being more generally (North 1990; Becker 1981; Soto 2003).

In contrast, a substantial body of theory and evidence alternatively suggests that formally legislated reforms, even when targeting inclusive institutional types that are typically correlated with "good" outcomes, do not necessarily cause or guarantee economic or social development. The relationship between institutions and social progress is not merely a set of "good" institutional ingredients. Instead, social and economic development are better thought of as a complex recipe that requires matching inclusive institutions on the one hand, with some form(s) of local socio-cultural conditions that happen to be hospitable to and appreciative of those economic institutional systems and processes on the other. If a society is culturally hostile to property rights and trade, and opposed to inclusive access to commercial life; then imposed legislation that formalizes inclusivity may fail to sustain as a lasting policy change, may carry limited economic returns, or even produce negative social outcomes.

This matching dynamic thus implies that when good institutional reforms sustain over time, they are typically lagging indicators of social processes already underway in the informal sector rather than original catalysts for additional social progress. Perhaps most notably, child labor laws reflect this lagging dynamic. As Nardinelli (1980), Snowden (1998), and Moehling (1999, 2002, 2004) have demonstrated, exogenous factors of technological change, economic development, changing birthrates, industrialism, economic opportunities, and material abundance had systematically reshaped dispersed individual and household choices away from child labor and toward enhanced education; thus, significantly reducing the number of children in the work force and improving material working conditions prior to formal labor regulations. The key insight here is that *de jure* legal institutions often codify rights and practices that have already occurred and even become stable norms through *de facto* behaviors and cultural processes (Williamson 2009). Furthermore, the measurable outcomes in terms of social well-being and women's material welfare are expected to exhibit diminishing returns over time. Thus, the earliest and greatest gains are more likely to have accrued via informal *de facto* behaviors in the past, the longer the exclusionary policy persisted through history.

Suppliers and demanders of any good or service, as well as any individual seeking to participate in some productive collective action, all have a strong vested interest not only in the strategic efficiency of their private choices, but also in the efficient operation of their respective social institutional policies and procedural norms. Institutions, simplistically defined as rules and enforcements (North 1990), can take form as either formal laws and public policies, or as informal social norms and cultural patterns of behavior. When choosing across formal and informal avenues to pursue individual plans like trade, entrepreneurship, or collective actions; rational agents will opt more toward those avenues that most maximize their private returns, reduce transaction costs, and mitigate against uncertainty. Hence, formal and informal institutions can serve as substitute pathways for accomplishing economic goals and social endeavors. When one avenue is particularly absent or obstructed, individuals will either chose to work for formal institutional change or seek viable

alternative pathways when affordable. Any situation or pattern of relative cost differences across formal and informal opportunities will tend toward some equilibrium ratio of choices and resultant pattern of institutional forms.

Prohibited activity is an obvious case in point. Lacking access to formal contracts, legal adjudication services or enforceable property rights to manage the stable and efficient production, sale, purchase, possession, transportation, or consumption of illicit products does not eliminate such preferences or behaviors from society. Rather, those willing participants who assess the likely benefits of trade to exceed the probabilistic potentials of costly sanctions, turn to the black-market rather than go without. In turn, black-market processes are not only positive and stable realities, they also tend to contain unique modes of securing and protecting property alongside distinctive cultural norms for enforcing contracts and curating trust for repeated dealings. As Hasnas (2024) has recently explained, the logical and historical foundations for the common law tradition suggest that though markets contexts vary tremendously in their size, specialization, historical and cultural contexts, all share an internal tendency to evolve specialized legal rules aimed at minimizing costly conflict, assuring capital productivity, and promoting relative economic performance given local constraints, technological capacities, and limited knowledge.

In the context of restricted economic rights for women, much like prohibition markets, women represent a profitable source of human capital and economic participation regardless of their formal legal status. Hence, women and willing trading partners will seek alternative means via informal institutional processes to perform extra-legal economic and social transactions even when the *de jure* laws exclude female participation. The likelihood of women being active participants in extra-legal institutions when lacking formal economic rights is likely greater than the tendency of illicit traders amidst traditional product prohibitions for three reasons. First, women represent a broader scope of economically productive potential than particular prohibited items or makers thereof. Second, exclusionary legal policies against women are not typically enforced with similar policing tactics or magnitudes as product prohibitions. Last, female exclusion is an incredibly long-standing cultural norm where it is practiced.

Extra legal economic and social participation by and with women will occur so long as their associated rewards exceed the probabilistically discounted rate of sanctioning. Again, like black market activities, both sides of such transactions have incentives to participate and profit in the extra-legal processes, and strong incentives against reporting or sanctioning their trading partners or similar trading agents throughout their community. Hence, there's a strong tendency for informal female participation rates to be greater than zero, if not substantially high, when *de jure* economic rights are absent. To assume such societies are effectively suppressing the totality of female economic and social productivity is to assume a significant degree of irrationality on the part of institutional authorities (male in such contexts) and citizen agents (both male and female) alike.

Historically, informal participation has most often proven to be the case, as women and their families developed institutional

mechanisms to perform and secure economic participation even before achieving legal rights. Consider one well-researched and notable example: under common law coverture, wealthy families deployed civically enforced contractual trusts to create separate estates for married women (Siegel 1993, 2018). If an elderly father possessed significant wealth, and wanted to assure his daughter's inheritance, independent from her husband's financial control; the trust could assure that the daughter had autonomy over the spending and titling of the relevant wealth. Trusts were also insulated from taxation relative to salaried income or common forms of propertied wealth formally held by men. These arrangements protected familial wealth transmission through female lines, and effectively secured *de facto* property rights for women, despite *de jure* legal systems denying equal economic participation.

This pattern extends beyond instruments to shape inheritance and instead encompasses a wide range of diverse economic and social arrangements. Women frequently create and maintain informal networks and governance structures that enhance their economic capabilities (Ostrom 1994). Empirical research in developing economies demonstrates how women establish sophisticated informal financial and trading networks when needed, even in the face of weak, absent, or oppositional legal rights (Chamlee-Wright 1997, 2002). In fact, these self-organizing systems often outperform their formal institutional correlates (Fafchamps 2003).

Market forces play a particularly crucial role in this process. When labor scarcity or economic necessity create uniquely high demands for women's participation, businesses and local economies often extend *de facto* rights and opportunities even when *de jure* rights are absent (Lemke 2016; Kramer 2024). Recent empirical work demonstrates this pattern across diverse contexts, from piece-rate compensation in textile manufacturing (Burnette 2024) to gender wage gaps among skilled workers (Burnette and Stanfors 2020). Historical evidence shows that firms limiting women's economic participation often face competitive disadvantages (Wolcott and Clark 1999), creating pressure for inclusive institutional change through market mechanisms rather than legal reforms.

Informal institutional channels such as social networks and family connections can and often do serve as crucial mechanisms for economic advancement. Historical evidence shows how women used family and community connections to access capital and develop business opportunities despite not having formal economic rights at the time (Lamoreaux 1986). In contemporary developing economies, women utilize similar strategies. With time and repetition, informal institutional norms facilitate and regularize *de facto* market participation despite *de jure* barriers (Meagher 2010).

This theory of institutional lag suggests that formal legal rights follow from rather than necessarily lead to economic and social changes (Fernández 2014; Williamson 2009). This lag can be explained by a variety of contributing factors. First, a basic application of diminishing returns helps explain the delay in formal institutions following informal female economic and social participation. Agents within a mixed context, of both formal and informal institutional opportunities, will

not only seek the lowest cost and highest yield routes for fulfilling their plans, but they will also seek the highest returns to their earliest strategic decisions of institutional navigation and institutional investments. If one is committed to trading in the extra-legal realm for lack of formal institutional inclusion, their earliest trades will likely entail, for example, selling their most profitable items sooner than those with lower potentials. The same would hold true for investing resources in institutional development or maintenance. When faced with the opportunity to spend costs and energy on building new relationships and norms, the highest return connections will be prioritized first. Hence, over time, additional actions and investments will likely yield lower marginal revenues.

Second, the institutional lag also occurs partly because existing power structures resist formal institutional changes even after informal adaptations have emerged (Doepke et al. 2012). However, economic pressures eventually necessitate formal recognition of women's rights as maintaining restrictions becomes increasingly costly (Duflo 2012). This helps explain why empirical analysis finds limited immediate effects from formal legal changes—the bulk of economic and social adjustments that drive improved outcomes have likely already occurred through informal channels.

Property rights theory provides additional insight into this pattern. As economies develop, the economic value of securing property rights for all productive members of society increases (Besley and Ghatak 2010). This creates pressure for institutional change, notably when excluding women from property ownership becomes increasingly costly. However, formally recognizing these rights through legal changes may codify existing practices rather than create new economic opportunities. Furthermore, observing a lack of formal institutional change may imply that such potentials are perceived as unlikely, not cost effective, or that satisfactory informal alternatives are available and being used.

Family structures also evolve in response to changing economic incentives (Horwitz 2015), often before de jure legal recognition catches up. Research shows that even within households, relative human capital differences between spouses drive changes in labor allocation and gender roles independent of formal legal structures (Taveras 2022). The finding that legal changes do not consistently improve social outcomes aligns with this framework—family adaptation to economic realities may be more important than formal rights in determining outcomes like education and health.

This theoretical framework explains our empirical findings by highlighting how women's advancement often proceeds through informal channels despite de jure legal barriers (Davis and Williamson 2022). Rather than waiting for legal reform, women historically have leveraged market mechanisms and informal institutions to enhance their economic and social position (Goldin 1991). The eventual legal recognition of women's rights through de jure change represents the culmination rather than the initiation of this process (Goldin 2006). When measuring the impact of legal changes, we may be capturing the tail end of a much longer process of institutional evolution (Goldin 1994).

This perspective suggests that controls—including education, government consumption, exports, inflation, democracy, and economic freedom—may capture the underlying mechanisms that truly drive improvements in women's status. The formal legal changes measured may thus appear ineffective not because rights do not matter, but because they typically formalize existing practices rather than create new opportunities (Kramer 2024). This insight carries significant implications for development policy, suggesting that supporting informal institutions and generalized economic prosperity may be more effective than focusing solely on legal reforms aimed at gender progress or equity.

Furthermore, this framework helps to recognize a precarious aspect of de jure institutional reform. In short, fast changes to inclusive formal institutions can represent major shocks to the relative strategies and institutional opportunities that agents choose and face. Thus, reducing the costs and increasing the opportunism of de jure institutional channels may very well reduce private investments and energies previously allocated to de facto informal processes, in turn weakening the participation rates and benefits associated with the extra legal market. Here we see institutions enduring a creative destruction dynamic akin to technology markets and or a pseudo-j-curve path of institutional transition. When switching toward a new institutional process, the old institutional processes will decline in use and measurable returns.

Returning to the context of changing toward gender inclusive economic rights, it is not the case that women increased economic participation from a starting point of zero. Instead, they were previously engaged in some equilibrium combination of un-enforced formal participation alongside some degree of extra-legal participation. Again, eliminating the sanctioning against formal participation does not guarantee greater amounts of productivity or profits on net. In fact, such effects on net can conceptually be negative.

First, newly allowed formal economic opportunities may represent and reflect greater amounts of uncertainty in the minds of women who were reliably securing profit through informal channels previously. Second, as suggested by Bastos and Pavlik (2024) surveyed above, formal participation may subject new female economic participants to disproportionately enforced formal regulations and discretionary scrutiny otherwise previously ignored and or successfully skirted in the informal extra-legal market. Last, newly inclusive formal processes may inspire new rates of informal sanctioning amongst those sub-populations who were opposed to and or overtly hostile to gender inclusivity (Jalota and Ho 2024) and or hostile to exogenous pressures for legal and institutional change (Boettke et al. 2015). Hence, our framework of institutional matching and policy lags accounts not only for the absence of positive effects from formal institutional changes, but also it helps to account for the instances of modest negative social welfare outcomes on the margin.

In summary, our theory is an explicitly economic rather than political or cultural approach to institutional change. As such, our first assumptions about reality must be dual commitments to individual (if only contextual) rationality and general (if

only limited) efficiency (Leeson 2020). An observed stable institutional norm, even one as illiberal and seemingly oppressive as female economic exclusion, in so far as it is long standing and internally stable, is representative of the agents within that system optimizing as best they can given limited opportunities. Hence, such norms conform to some degree of efficiency (Stigler 1992).

Our theoretical framework should not be interpreted to suggest that women's legal rights don't matter. We would like to highlight a socially optimistic interpretation of our argument instead. First, women who lived and currently exist in oppressive legal environments are not passive but active agents discovering and producing value despite formal legal obstacles. Second, any regime, even those with deep forms of gendered unfairness can discover informal and or extra legal means to obtain comparable economic returns as would be viable under greater rates of formal gender equality.

5 | Data

5.1 | Economic Freedom of the World

The Economic Freedom of the World (EFW) index has become a widely used measure of economic liberty across nations (Gwartney 2009). This index quantifies how a country's institutions and policies align with personal choice, voluntary exchange, open markets, and clearly defined property rights using objective data and survey information (Gwartney and Lawson 2003). Comprising 42 distinct components currently available for 141 countries, the EFW index facilitates empirical investigations into the institutional theory of growth and enables cross-country comparisons of economic performance (Gwartney 2009).

The literature surrounding economic freedom has grown substantially since the index's inception, with numerous studies exploring its relationship to various social outcomes. Hall and Lawson (2014) conducted a comprehensive review of 402 articles citing the EFW index, finding that 198 used it as an independent variable in empirical studies. Notably, over two-thirds of these studies reported a positive association between economic freedom and desirable outcomes such as faster growth, improved living standards, and increased happiness.

One key area of research has been the investigation of the relationship between economic freedom and economic growth. Faria and Montesinos (2009) employed instrumental variable (IV) methods to address endogeneity concerns and uncovered a robust causal channel from economic freedom to prosperity. Their findings support the importance of policies and institutions in fostering development and provide a basis for policy recommendations to increase economic freedom.

Researchers have also explored the interplay between economic freedom and political systems. Rode and Gwartney (2012) examined the impact of transitions to democracy upon economic freedom, finding that such transitions are generally associated with subsequent increases in economic liberalization. Their study revealed an inverted U-shaped pattern of economic liberalization

following democratic transitions, with liberalization increasing for approximately 10 years before receding.

The role of international institutions in promoting economic freedom has been another area of inquiry. Boockmann and Dreher (2003) analyzed the effects of World Bank and IMF policies on the EFW index and its sub-indexes. They found that the number of World Bank projects had a positive impact on overall economic freedom, particularly in the 1990s. At the same time, the amount of World Bank credits appeared to have a negative effect. However, they found no clear relationship between IMF credits or programs and economic freedom as measured by the index.

Despite the generally positive findings associated with economic freedom, challenges remain in identifying which specific aspects of economic freedom are most conducive to growth. Rode and Coll (2012) use cluster analysis to reorganize the Economic Freedom of the World Index components into seven clusters, with three main multi-variable clusters showing distinct patterns. Cluster 2, which combined legal structure, security of property rights, and business regulations, demonstrated significant long-term growth effects when measured in levels, suggesting that institutional credibility and consistent property rights protection are fundamental for sustained economic growth. Cluster 3, which merged monetary issues with most foreign trade variables, showed no significant direct effect on growth, though the authors note this could potentially operate through indirect channels. The separate variable clusters for government enterprises/investment, capital controls, foreign currency bank accounts, and conscription generally showed no robust relationship with growth, with only conscription showing some significant but non-robust effects.

Cluster 1, which combined government size with labor market regulations, showed a different pattern—it had significant positive effects on growth only when measuring changes rather than levels. This suggests that reducing government size or liberalizing labor markets can provide immediate growth benefits, but these effects may be primarily short-term. Notably, the analysis revealed that variables related to regulation split differently than in the original index—credit market regulations joined monetary variables, business regulations aligned with legal structure, and labor market regulations grouped with government size. This reorganization helped resolve previous multicollinearity issues and demonstrated why earlier studies had difficulty isolating specific effects, as the growth impacts operated through different mechanisms and over varying time horizons depending on the specific aspect of economic freedom being measured. This paper recalculates the Economic Freedom of the World Index aggregate metric by removing the Gender Disparity Index (GDI) from the analysis. This is necessary because our treatment is a change in the GDI.

5.2 | Gender Disparity Index

The Gender Disparity Index (GDI) is a measure developed to quantify the degree of de jure legal equality between men and women across countries (Fike 2016). The core economic

rights measured in the Gender Disparity Index encompass the fundamental elements required for women's inclusive participation in the formal economy. Property rights form a crucial foundation, including not only the ability to own immovable property but also to inherit wealth both as daughters from parents and as spouses, placing women on equal legal footing with their male counterparts in wealth accumulation and transfer. Professional rights ensure that women can pursue any occupation or trade without requiring permission from male relatives, while also guaranteeing their ability to work night shifts and in industries traditionally restricted to men, such as mining, construction, and manufacturing. Financial autonomy is protected through rights that allow women to independently manage their economic affairs, including opening and operating bank accounts, entering legally binding contracts, and conducting business transactions without requiring a male guarantor. The freedom to establish and operate businesses extends to women's right to register companies, obtain necessary licenses, and run enterprises without additional gender-based restrictions. Mobility rights guarantee that women can travel both within and outside their country, apply for passports, and choose their place of residence without male permission. Legal capacity rounds out these economic freedoms by ensuring that women can initiate court proceedings, serve as heads of household, and make independent legal decisions that affect their economic interests, establishing them as fully autonomous economic actors under the law.

It is constructed using data from the World Bank's Women, Business, and the Law Report, the GDI is based on 17 yes/no questions across seven categories of rights. Scores range from 0 to 1, with 1 indicating complete legal equality and 0 representing disparity in all measured areas. The GDI focuses solely on formal legal and regulatory codes, excluding informal factors like social norms or religious customs. Its primary application is in adjusting economic freedom indices, particularly the Economic Freedom of the World (EFW) index, to account for gender-based legal disparities. This adjustment is most notably applied to the Legal System and Property Rights area of the EFW, using the formula:

$$\text{Adjusted Area 2 Score}_{it} = \frac{\text{Area 2 Score}_{it} + \text{GDI Adjustment}_{it} * \text{Area 2 Score}_{it}}{2}$$

This equation effectively averages the unadjusted and adjusted scores, reflecting the impact of gender disparity on approximately half the population. The GDI's importance lies in its ability to highlight and quantify legal gender inequalities, track changes over time (from 1970 to 2021), and provide a more comprehensive view of a country's economic environment. By incorporating gender equality into assessments of economic freedom, the GDI offers valuable insights for researchers and policy-makers, potentially influencing country rankings and policy decisions. Ultimately, the GDI is a crucial tool in understanding the relationship between legal gender equality and various economic outcomes, emphasizing equal rights in fostering economic development and freedom for all individuals in society.

The GDI data spans 1970 to 2021, with historical data reconstructed for earlier years. Over this period, the global average

GDI score increased from 0.68 in 1970 to 0.87 in 2021, indicating a general trend toward greater gender equality under the law. As Table 1 shows, countries in the Middle East and North Africa tend to have lower GDI scores. In contrast, many developed and some developing nations have achieved perfect

TABLE 1 | Gender Disparity Index for 2019.

ISO	GDI	ISO	GDI	ISO	GDI	ISO	GDI
AGO	0.88	ALB	1.00	ARE	0.59	ARM	1.00
AUS	1.00	AUT	1.00	BDI	0.82	BEL	1.00
BEN	0.76	BFA	0.82	BGD	0.76	BGR	1.00
BHR	0.47	BOL	1.00	BRA	1.00	BRB	0.88
BWA	0.88	CAF	0.76	CAN	1.00	CHE	1.00
CHL	0.82	CHN	0.88	CIV	0.88	CMR	0.47
COG	0.59	COL	0.94	CRI	0.88	CYP	0.88
CZE	1.00	DEU	1.00	DNK	1.00	DOM	1.00
DZA	0.71	ECU	1.00	EGY	0.41	ESP	1.00
EST	1.00	ETH	0.88	FIN	1.00	FJI	0.88
FRA	1.00	GAB	0.41	GBR	1.00	GHA	0.94
GMB	0.88	GRC	1.00	GTM	0.94	GUY	0.94
HKG	1.00	HND	0.94	HRV	1.00	HTI	0.82
HUN	1.00	IDN	0.82	IND	0.88	IRL	1.00
IRN	0.41	IRQ	0.47	ISL	1.00	ISR	0.88
ITA	1.00	JAM	0.94	JOR	0.41	JPN	0.94
KAZ	0.88	KEN	0.88	KGZ	0.88	KHM	1.00
KOR	0.88	KWT	0.47	LAO	1.00	LBR	1.00
LKA	0.88	LSO	0.94	LTU	1.00	LUX	1.00
LVA	1.00	MAR	0.76	MDA	1.00	MDG	0.76
MEX	1.00	MLI	0.65	MLT	0.94	MMR	0.88
MNG	1.00	MOZ	0.94	MRT	0.47	MUS	1.00
MWI	0.88	MYS	0.65	NAM	0.94	NER	0.53
NGA	0.82	NIC	1.00	NLD	1.00	NOR	1.00
NPL	0.94	NZL	1.00	PAK	0.65	PAN	0.94
PER	1.00	PHL	0.82	POL	1.00	PRT	1.00
PRY	1.00	QAT	0.53	ROU	1.00	RUS	0.88
RWA	1.00	SAU	0.71	SDN	0.29	SEN	0.65
SGP	1.00	SLE	0.94	SLV	1.00	SRB	1.00
SVK	1.00	SVN	0.94	SWE	1.00	SWZ	0.53
TGO	1.00	THA	0.94	TTO	0.94	TUN	0.71
TUR	0.94	TZA	0.88	UGA	0.82	UKR	0.82
URY	0.94	USA	1.00	VNM	0.88	ZAF	1.00
ZMB	0.94	ZWE	1.00				

scores of 1.00. Notably, countries with higher economic freedom tend to have higher GDI scores, suggesting a positive relationship between economic freedom and gender equality under the law.

The GDI is used to adjust Area 2 (Legal System and Property Rights) of the EFW index. This adjustment significantly impacts the scores and rankings of some countries, particularly in the MENA region. However, it's important to note that the GDI is a *de jure* measure based on formal laws and regulations. It does not capture informal institutions, social norms, or the actual enforcement of laws. This suggests that this framework could be expanded to incorporate equality measures for other marginalized groups as more data becomes available. The introduction of the GDI into the EFW index represents an effort to more accurately measure economic freedom by considering the legal rights and economic opportunities available to all members of society, as opposed to just men in the countries lacking legal gendered equality.

Our data defines a treatment of countries that experienced at least a 0.15 increase out of 0 to 1 change in GDI that persisted for 5 years. We also run the same comparison for a 0.125 increase and a 0.1 increase as a robustness check. The only reversal observed in the data is Haiti's GDI score in 2021, which was lower than in 2020. This reversal is minor and affects only the final year of our dataset, making it unlikely to bias our results significantly.

5.3 | Outcomes and Covariates

The Penn World Table (PWT) (Feenstra et al. 2015) provides our measure of real GDP per capita (*rgdpe_per_capita*). This variable, derived from the *rgdpe* measure in PWT version 10.01, offers a comprehensive view of economic output across countries and over time, adjusted for purchasing power parity. We also use their measure of human capital, which combines average years of schooling with returns to education, using a piecewise linear function that accounts for diminishing returns at higher education levels. This measurement, denoted as "hc" in the database, converts educational attainment into an effective measure of human capital by considering both the quantity of schooling and how it translates into workforce productivity.

Most of our socioeconomic indicators are sourced from the World Bank's World Development Indicators (WDI). This extensive database contributes variables such as maternal mortality ratios, urban sanitation access, and life expectancy. Additionally, the WDI supplies our macroeconomic control variables: government consumption expenditure, exports as a percentage of GDP, and inflation rates. These indicators provide a multifaceted view of human capital development, public health, and macroeconomic conditions.

Lastly, our measures of democratic quality are drawn from the Varieties of Democracy (V-Dem) database (Coppedge et al. 2024). The polyarchy index (*v2x_polyarchy*) is our primary indicator of democratic development.

6 | Method

We use Propensity Score Matching (PSM), Mahalanobis Distance Matching (MDM), and Callaway and Sant'Anna (2021) as robustness checks to estimate the effects. These methods are valuable for addressing selection bias and endogeneity issues often arising in cross-country studies.

PSM, as utilized by Callais and Young (2023) in their study of economic freedom and inequality, involves a two-step process. First, it estimates a unit's probability (propensity score) receiving treatment based on observed characteristics. Then, it matches treated units with non-treated units having similar propensity scores. This method enables researchers to create a quasi-experimental setting using observational data, thereby balancing the distribution of observed covariates between the treated and control groups.

MDM matches treated units to control units directly based on the similarity of their observed covariates. Pavlik and Callais (2024) employed both PSM and MDM in their analysis of corruption and income inequality, demonstrating the complementary nature of these techniques. MDM calculates the Mahalanobis distance, which accounts for correlations between variables, to find control units most similar to each treated unit across all observed covariates simultaneously.

Both methods are instrumental in studying the Economic Freedom of the World (EFW) index. For instance, Grier (2023) used these techniques to examine the causal effects of significant increases in economic freedom on women's labor force participation and education. She found that significant increases in economic freedom positively impact female labor participation and primary education completion. The results show modest positive effects on women's labor participation but more substantial improvements in female primary school completion rates. Similarly, Alvarez et al. (2024) applied matching methods to revisit the relationship between economic freedom and development, accounting for potential statistical deception by autocratic regimes.

These matching methods allow researchers to estimate average treatment effects by comparing outcomes in treated countries (those experiencing significant changes in economic freedom) to similar but non-treated countries. As demonstrated by Lawson (2024) in studying economic freedom and fertility rates, these techniques can provide robust insights into the complex relationships between economic policies and various socioeconomic outcomes.

By using both PSM and MDM, researchers can test the robustness of their findings, as each method has its strengths depending on the specific characteristics of the data and research question at hand. This approach has proven valuable in advancing our understanding of the impacts of economic freedom across various dimensions of social and economic development.

As a robustness check, we employ the difference-in-differences methodology developed by Callaway and Sant'Anna (2021) to

conduct an event study analysis of our key outcome variables, using a 0.1 change in GDI over 5 years as our treatment threshold. We implement this analysis using the *did* package in R (Callaway and Sant'Anna 2021), which provides doubly robust staggered treatment difference-in-differences estimation. This methodological approach is particularly valuable as it addresses fundamental challenges inherent in settings with staggered treatment adoption and heterogeneous treatment effects across units and time.

Our empirical strategy defines treatment cohorts based on the timing of substantial changes in the Gender Disparity Index, with countries entering the treatment group upon experiencing a five-year increase in GDI of 0.1 or greater. The Callaway-Sant'Anna estimator computes group-time average treatment effects (ATT(g,t)) for each treatment cohort and time period combination, utilizing never-treated units as the control group. This approach effectively circumvents the methodological concerns associated with two-way fixed effects models, particularly the issues of implicit weighting schemes and potential negative weights. The implementation accommodates unbalanced panels and employs doubly-robust estimation, integrating both outcome regression and propensity score weighting to strengthen the analysis against model misspecification.

We extend our analysis to examine dynamic treatment effects through the dynamic *aggte* function, which aggregates the group-time ATTs into event-study estimates. This aggregation preserves the interpretation of the underlying group-time ATTs

while accounting for treatment effect heterogeneity across cohorts. Importantly, this methodology prevents the contamination of dynamic treatment effects often occurring in traditional two-way fixed effects approaches when treatment effects exhibit temporal variation.

7 | Results

We use Propensity Score Matching and Mahalanobis Distance Matching to identify any potential biases in the presented analysis. Propensity Score Matching (PSM) and Mahalanobis Distance Matching (MDM) are particularly well-suited for examining the effect of increased gender development on various outcome markers, as they effectively address selection bias. These methods enable a more robust estimation of the effects by matching treated units (countries experiencing significant increases in gender development) with similar non-treated units based on observed characteristics, thereby mitigating potential confounding factors and providing a clearer picture of the relationship between gender development and socioeconomic outcomes.

Table 2, Table 3, and Figure 1a provide insight into the effectiveness of Propensity Score Matching (PSM) in balancing the treated and control groups for the analysis of the impact of gender development on various outcomes. The tables present balance statistics before and after matching, which are crucial for assessing the quality of the matching process.

TABLE 2 | PSM balance before matching.

	Means treated	Means control	Std. mean diff.	Var. ratio	eCDF mean	eCDF max
distance	0.05	0.04	0.43	0.98	0.13	0.23
lag_hc	2.11	2.35	−0.40	0.73	0.10	0.20
lag_gov_consumption	0.17	0.18	−0.19	0.58	0.03	0.06
lag_exports	0.19	0.26	−0.46	0.30	0.07	0.15
lag_inflation	37.59	25.30	0.05	0.33	0.06	0.14
lag_democracy	0.53	0.55	−0.06	0.92	0.03	0.11
lag_efw	6.21	6.48	−0.23	0.80	0.07	0.17

TABLE 3 | PSM balance after matching.

	Means treated	Means control	Std. mean diff.	Var. ratio	eCDF mean	eCDF max	Std. pair dist.
distance	0.05	0.05	0.00	1.00	0.00	0.01	0.00
lag_hc	2.11	2.12	−0.03	0.82	0.03	0.10	0.72
lag_gov_consumption	0.17	0.16	0.12	0.88	0.03	0.08	1.07
lag_exports	0.19	0.20	−0.07	0.59	0.03	0.10	0.83
lag_inflation	37.59	25.95	0.05	2.67	0.03	0.08	0.22
lag_democracy	0.53	0.54	−0.03	0.92	0.03	0.08	1.05
lag_efw	6.21	6.18	0.03	0.71	0.04	0.10	1.06

Before matching, there were substantial differences between the treated and control groups across several key variables, as indicated by the relatively large standardized mean differences and variance ratios that deviate from 1. For instance, variables such as lag_hc (lagged human capital), lag_exports, and the distance (propensity score) themselves showed considerable imbalance, with standardized mean differences of -0.40 , -0.46 , and 0.43 , respectively. This imbalance suggests that the treated and control groups were not directly comparable before matching, which could lead to biased estimates of the treatment effect.

After matching, the balance between the treated and control groups improved significantly across most variables. The standardized mean differences for all variables decreased substantially, most falling below the standard threshold of 0.1 in absolute value. For example, the imbalance in lag_hc reduced from -0.40 to -0.03 , and for lag_exports, from -0.46 to -0.07 . The variance ratios also generally moved closer to 1, indicating improved balance in the distribution of these variables between the treated and control groups. This improvement in balance suggests that the matching process was largely successful in creating comparable groups, thereby reducing potential bias in estimating treatment effects. This can also be seen in Figure 1.

The variables being controlled for in this analysis cover a wide range of economic, social, and political factors. These include measures of human capital (school enrollment rates, average years of education), macroeconomic conditions (government consumption, exports, inflation), political institutions

(democracy index), and economic freedom. By controlling for these factors, the analysis aims to isolate the effect of improvements in gender development from other potential confounding variables that could influence the outcomes of interest. This comprehensive set of controls enhances the credibility of the claims made in the study by accounting for major alternative explanations for changes in the outcome variables.

Table 4 and Table 5 present the balance statistics for the Mahalanobis Distance Matching (MDM) process, offering insight into the effectiveness of this matching technique in creating comparable treated and control groups for the analysis of gender development impacts.

Before matching, substantial imbalances existed between the treated and control groups across several key variables. The standardized mean differences reveal large disparities in lagged human capital (lag_hc) and lagged exports (lag_exports), with values of -0.40 and -0.46 , respectively. Additionally, the variance ratios for most variables deviated considerably from 1, especially for lag_exports (0.30) and lag_inflation (0.33), indicating significant differences in the distribution of these variables between the groups. These imbalances suggest that before matching, the treated and control groups were not directly comparable, potentially leading to biased estimates of treatment effects.

After applying Mahalanobis Distance Matching, the balance between the treated and control groups improved dramatically

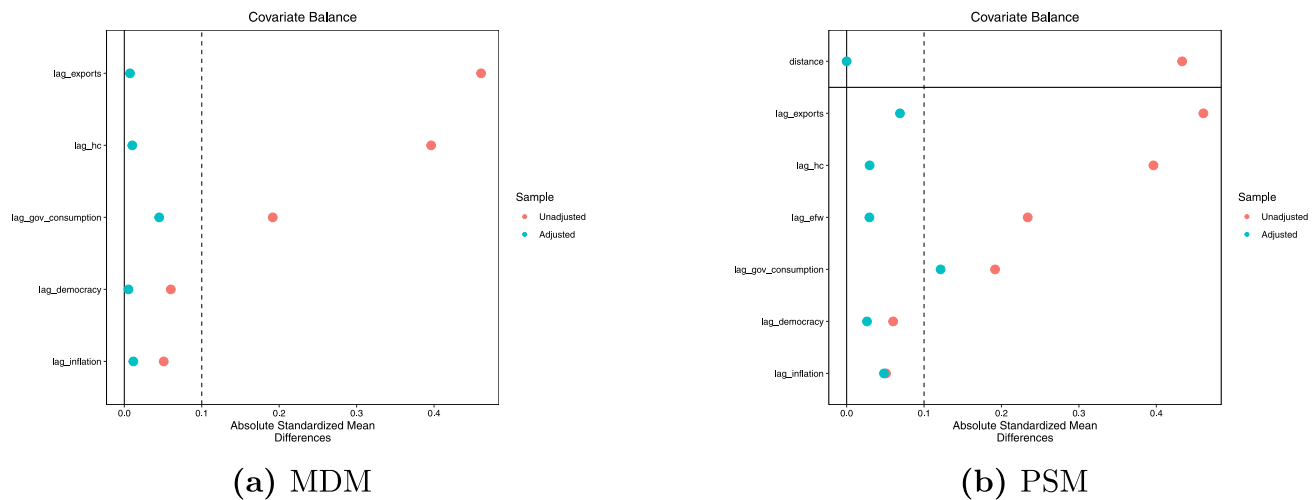


FIGURE 1 | Comparison of MDM and PSM love plots. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

TABLE 4 | MDM balance before matching.

	Means treated	Means control	Std. mean diff.	Var. ratio	eCDF mean	eCDF max
lag_hc	2.11	2.35	-0.40	0.73	0.10	0.20
lag_gov_consumption	0.17	0.18	-0.19	0.58	0.03	0.06
lag_exports	0.19	0.26	-0.46	0.30	0.07	0.15
lag_inflation	37.59	25.30	0.05	0.33	0.06	0.14
lag_democracy	0.53	0.55	-0.06	0.92	0.03	0.11

across all variables. The standardized mean differences for all variables decreased to near zero, with the largest absolute value being just 0.04 for lag_gov_consumption. Similarly, the variance ratios moved closer to 1 for all variables, with the most extreme case being lag_inflation at 0.88. The substantial improvements in these balance statistics indicate that the MDM process was highly effective in creating comparable groups. This enhanced balance suggests that the matched samples are now much more suitable for estimating the effects of gender development improvements, as the influence of these potentially confounding variables has mainly been equalized between the treated and control groups. The success of the MDM in achieving this balance lends credibility to the subsequent analysis of treatment effects, as it helps to isolate the impact of the treatment (improvements in gender development) from other factors that could influence the outcomes of interest.

Table 6 compares various matching methods used to estimate the effect of a significant increase in gender development (defined as a 0.15 jump in the Gender Development Index over

5 years) on several socioeconomic indicators. The methods compared include Propensity Score (PS) matching with different numbers of nearest neighbors, as well as Mahalanobis Distance Matching (MDM) with varying numbers of nearest neighbors.

The results in Table 6 show mild variation across different matching techniques, highlighting the sensitivity of the estimates to the choice of method. For most outcomes, the direction of the effect is consistent across methods, but the magnitude and statistical significance vary.

For GDP per capita (rgdpe_per_capita), most methods suggest a negative effect, with PS nearest neighbor methods showing statistically significant decreases ranging from about \$2100 to \$2200 dollars. However, MDM results are not significant and the results from these models are small.

Maternal mortality shows consistently positive and statistically significant effects across all methods, indicating an increase

TABLE 5 | MDM balance after matching.

	Means treated	Means control	Std. mean diff.	Var. ratio	eCDF mean	eCDF max	Std. pair dist.
lag_hc	2.11	2.11	−0.01	0.97	0.01	0.04	0.10
lag_gov_consumption	0.17	0.17	−0.05	1.03	0.02	0.07	0.14
lag_exports	0.19	0.19	0.01	0.99	0.02	0.08	0.15
lag_inflation	37.59	34.77	0.01	0.88	0.02	0.07	0.05
lag_democracy	0.53	0.53	−0.01	0.99	0.01	0.05	0.08

TABLE 6 | Matching methods comparison treatment 5yr. 0.15.

Method	rgdpe_per_capita	maternal_mortality	sanitation_urban	life_expectancy
PS: Nearest Neighbor	−2120.182* (1172.486)	114.226** (46.076)	−7.039 (4.311)	−2.275** (1.012)
PS: Nearest 2 Neighbors	−2169.822* (1172.486)	111.013** (46.076)	−6.9 (4.311)	−2.26** (1.012)
PS: Nearest 3 Neighbors	−2195.89* (1172.486)	111.63** (46.076)	−7.249 (4.311)	−2.182** (1.012)
PS: Nearest 4 Neighbors	−2217.909* (1172.486)	115.124** (46.076)	−6.985 (4.311)	−2.146** (1.012)
Mahalanobis: Nearest Neighbor	−537.267 (1066.694)	111.825*** (42.675)	−7.485* (4.138)	−1.492 (0.94)
Mahalanobis: Nearest 2 Neighbors	−651.064 (1066.694)	118.903*** (42.675)	−7.631* (4.138)	−1.52 (0.94)
Mahalanobis: Nearest 3 Neighbors	−596.075 (1066.694)	116.457*** (42.675)	−7.52* (4.138)	−1.501 (0.94)
Mahalanobis: Nearest 4 Neighbors	−624.458 (1066.694)	115.337*** (42.675)	−7.629* (4.138)	−1.513 (0.94)

in maternal mortality rates associated with the treatment. The magnitude of this effect varies, from about 111 (MDM) to 118 (PS nearest neighbor) additional deaths per 100,000 live births.

Urban sanitation access shows mixed results, with MDM methods indicating statistically significant adverse effects, while other methods show insignificant effects. Life expectancy

demonstrates a consistent pattern of small but statistically significant decreases for PSM and insignificant effects with MDM, ranging from about 1.5 to 2.3 years.

These results underscore the complexity of the relationship between gender development and various socioeconomic outcomes and the importance of methodological choices in

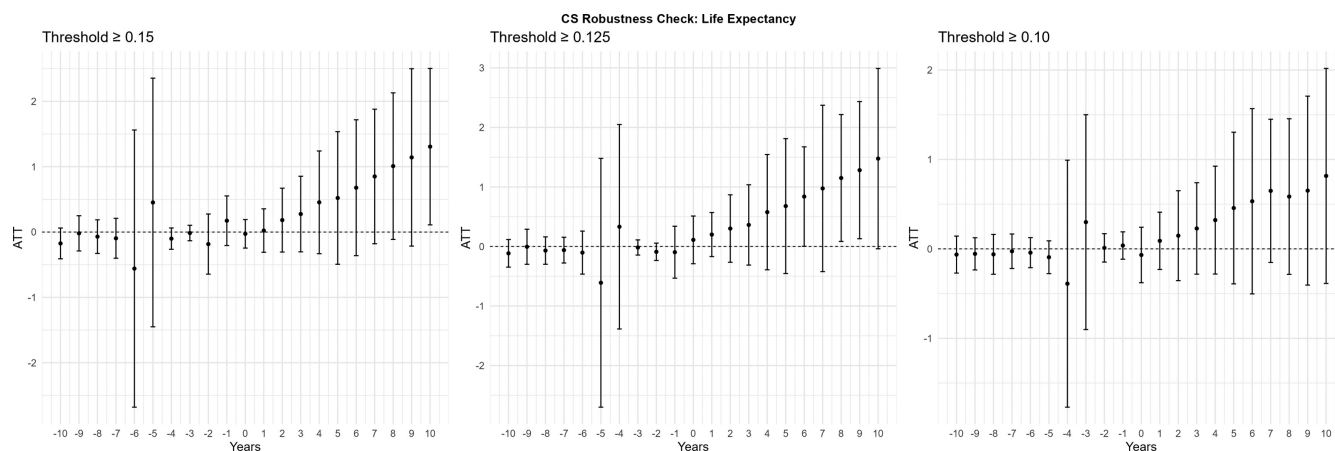


FIGURE 2 | Life expectancy.

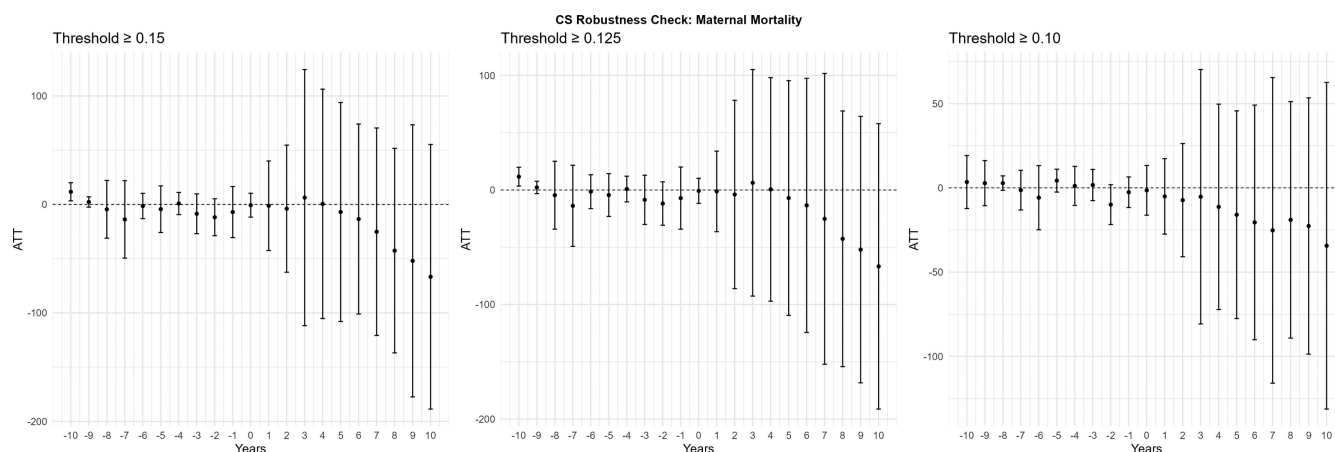


FIGURE 3 | Maternal mortality.

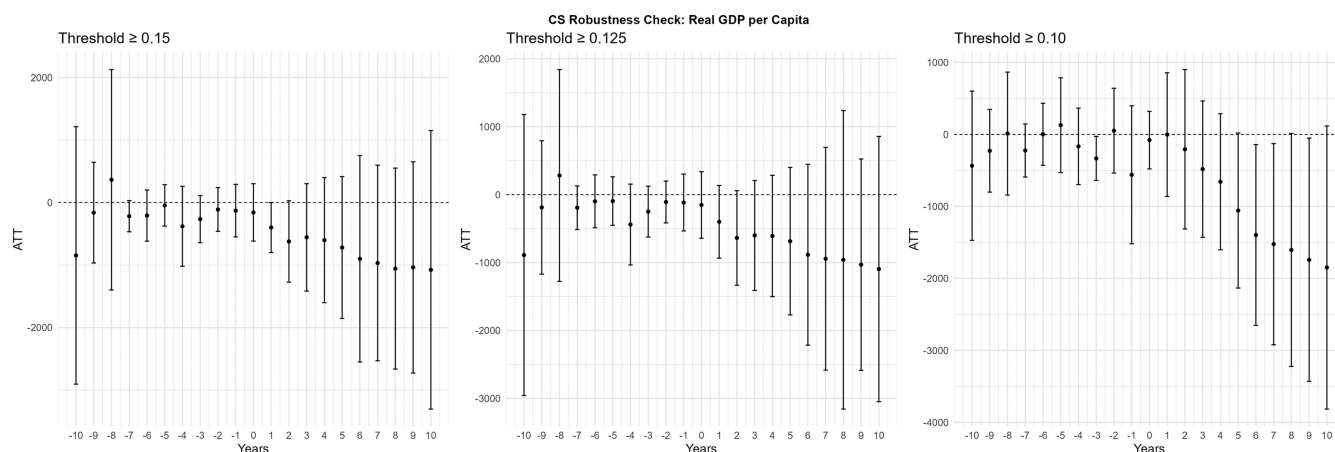


FIGURE 4 | GDP per capita.

estimating these effects. The inconsistencies across methods, suggest the need for careful interpretation and potentially further investigation into the underlying data structure and the appropriateness of different matching techniques for this specific analysis. There is a possibility that our treatment, an increase of 0.15 in GDI that persists over 5 years is driving the results by selecting an idiosyncratic sample. Therefore, we change the definition to address this concern. In the Appendix, we present these findings, however we do not find

statistically significant improvements in any columns in the tables for 0.125 or 0.15 jumps in GDI.

Finally, we use Callaway and Sant'Anna (2021). to ensure that our counter-intuitive results are not merely an artifact of an inappropriate empirical strategy. The resulting dynamic treatment effects are visualized in Figure 2 for life expectancy, Figure 3 for maternal mortality, Figure 4 for GDP per capita, and Figure 5 for urban sanitation. These results largely align with our findings in

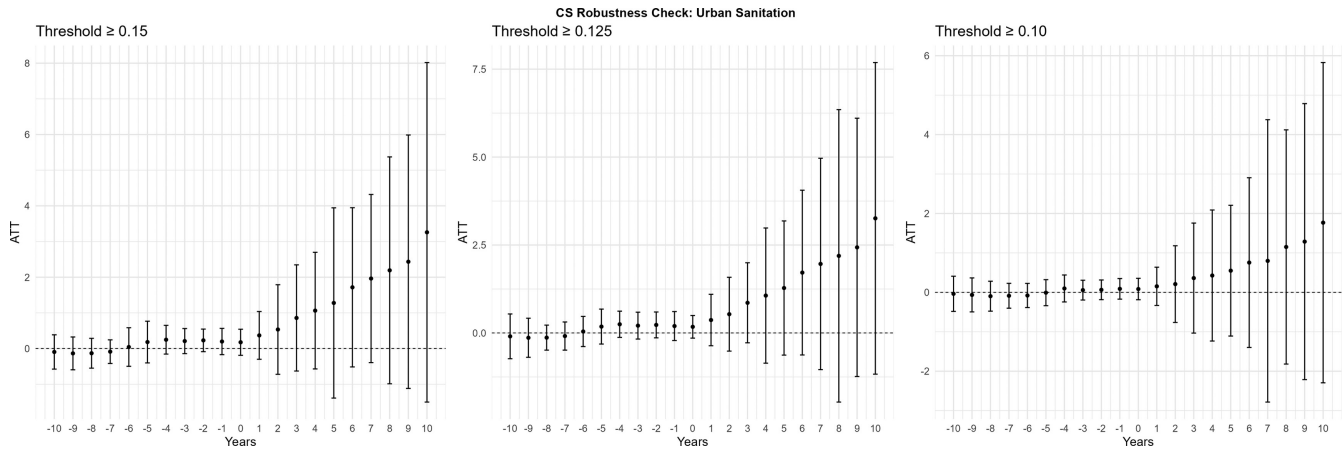


FIGURE 5 | Urban sanitation.

TABLE 7 | Callaway & Sant'Anna results summary.

	Real GDP per capita	Life expectancy	Maternal mortality	Urban sanitation
<i>Panel A: GDI Jump > = 0.1</i>				
ATT	-2687.50	0.56	-4.62	1.77
SE	(2029.78)	(0.36)	(9.87)	(1.29)
CI Lower	-6665.87	-0.15	-23.97	-0.76
CI Upper	1290.87	1.27	14.73	4.30
P-value	0.19	0.12	0.64	0.17
<i>Panel B: GDI Jump > = 0.125</i>				
ATT	-1096.68	0.27	0.98	0.62
SE	(2305.08)	(0.66)	(9.66)	(0.90)
CI Lower	-5614.64	-1.02	-17.95	-1.14
CI Upper	3421.28	1.56	19.91	2.38
P-value	0.63	0.69	0.92	0.50
<i>Panel C: GDI Jump > = 0.15</i>				
ATT	-1059.26	0.28	0.98	0.62
SE	(1944.92)	(0.75)	(8.31)	(0.97)
CI Lower	-4871.30	-1.19	-15.31	-1.28
CI Upper	2752.78	1.75	17.27	2.52
P-value	0.59	0.71	0.91	0.53

Note: Confidence intervals calculated at the 95% level using $ATT \pm 1.96 \times SE$.
Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 9. While we observe a modest positive trend in life expectancy and a decline in GDP per capita, the comprehensive results presented in Table 7 indicate that none of these effects achieve statistical significance under the Callaway and Sant'Anna (2021) framework. Additional analysis in Tables 7–22.

8 | Discussion and Implications

Our analysis suggests that de jure legal changes in women's rights, as measured by significant increases in the Gender Disparity Index, do not consistently produce the improvements

TABLE 8 | Matching methods comparison treatment 5yr 0.125.

Method	rgdpe_per_capita	maternal_mortality	sanitation_urban	life_expectancy
PS: Nearest Neighbor	−1039.94 (1159.446)	65.206 (45.755)	−0.776 (4.032)	−1.055 (0.947)
PS: Nearest 2 Neighbors	−998.656 (1159.446)	71.304 (45.755)	−0.891 (4.032)	−1.111 (0.947)
PS: Nearest 3 Neighbors	−1000.703 (1159.446)	69.017 (45.755)	−0.796 (4.032)	−1.087 (0.947)
PS: Nearest 4 Neighbors	−935.077 (1159.446)	68.246 (45.755)	−0.696 (4.032)	−0.984 (0.947)
Mahalanobis: Nearest Neighbor	−516.253 (967.156)	100.449** (44.888)	−6.407 (3.957)	−1.468 (0.909)
Mahalanobis: Nearest 2 Neighbors	−525.479 (967.156)	98.023** (44.888)	−6.283 (3.957)	−1.444 (0.909)
Mahalanobis: Nearest 3 Neighbors	−539.339 (967.156)	96.441** (44.888)	−6.549 (3.957)	−1.501 (0.909)
Mahalanobis: Nearest 4 Neighbors	−538.791 (967.156)	101.68** (44.888)	−6.142 (3.957)	−1.481 (0.909)

TABLE 9 | Matching methods comparison treatment 5yr 0.1.

Method	rgdpe_per_capita	maternal_mortality	sanitation_urban	life_expectancy
PS: Nearest Neighbor	−1300.683 (907.182)	55.485* (28.436)	−3.58 (2.835)	−0.343 (0.736)
PS: Nearest 2 Neighbors	−1391.484 (907.182)	57.18* (28.436)	−3.301 (2.835)	−0.306 (0.736)
PS: Nearest 3 Neighbors	−1351.604 (907.182)	54.931* (28.436)	−3.605 (2.835)	−0.185 (0.736)
PS: Nearest 4 Neighbors	−1395.415 (907.182)	53.206* (28.436)	−3.326 (2.835)	−0.344 (0.736)
Mahalanobis: Nearest Neighbor	−920.313 (821.993)	45.067 (28.484)	−2.162 (2.806)	−1.082 (0.7)
Mahalanobis: Nearest 2 Neighbors	−1022.138 (821.993)	42.597 (28.484)	−1.97 (2.806)	−1.082 (0.7)
Mahalanobis: Nearest 3 Neighbors	−985.51 (821.993)	44.843 (28.484)	−2.024 (2.806)	−1.085 (0.7)
Mahalanobis: Nearest 4 Neighbors	−945.528 (821.993)	42.993 (28.484)	−2.227 (2.806)	−1.067 (0.7)

TABLE 10 | Women, business, and the law data.

Economy	Economy code	Region	WBL 2024 EO 2.0 index	WBL 2024 Legal 2.0	EFW jump years treated
Algeria	DZA	Middle East and North Africa	82.5	40	2000–2005
Angola	AGO	Sub-Saharan Africa	66.25	62.5	1982–1987
Argentina	ARG	Latin America and Caribbean	56.25	75	1982–1987
Armenia	ARM	Europe and Central Asia	71.875	75	2000–2005
Australia	AUS	High income: OECD	88.75	90	1980–1985
Austria	AUT	High income: OECD	85.625	90	1972–1977
Bahamas, The	BHS	Latin America and Caribbean	63.75	55	1997–2002
Belgium	BEL	High income: OECD	90	90	1972–1977
Benin	BEN	Sub-Saharan Africa	65.625	70	2000–2005
Bolivia	BOL	Latin America and Caribbean	65	70	1970–1975
Botswana	BWA	Sub-Saharan Africa	57.5	52.5	2000–2005
Brazil	BRA	Latin America and Caribbean	61.875	80	1983–1988
Brunei Darussalam	BRN	East Asia and Pacific	91.25	35	2005–2010
Burkina Faso	BFA	Sub-Saharan Africa	62.5	65	1982–1987
Burundi	BDI	Sub-Saharan Africa	56.875	52.5	1988–1993
Cabo Verde	CPV	Sub-Saharan Africa	66.25	70	1976–1981
Central African Republic	CAF	Sub-Saharan Africa	35	52.5	1993–1998
Chile	CHL	High income: OECD	62.5	77.5	1982–1987
Colombia	COL	Latin America and Caribbean	63.75	77.5	1970–1975
Congo, Rep.	COG	Sub-Saharan Africa	46.25	45	1980–1985
Côte d'Ivoire	CIV	Sub-Saharan Africa	53.75	77.5	2008–2013
Dominican Republic	DOM	Latin America and Caribbean	67.5	72.5	1972–1977
Ecuador	ECU	Latin America and Caribbean	66.25	82.5	1970–1975
El Salvador	SLV	Latin America and Caribbean	31.25	82.5	1990–1995
Estonia	EST	High income: OECD	92.5	85	2004–2009
Eswatini	SWZ	Sub-Saharan Africa	40.625	35	2007–2012
Ethiopia	ETH	Sub-Saharan Africa	43.125	60	1996–2001
Fiji	FJI	East Asia and Pacific	30.625	62.5	2003–2008
France	FRA	High income: OECD	72.5	90	1983–1988

(Continues)

TABLE 10 | (Continued)

Economy	Economy code	Region	WBL 2024 EO 2.0 index	WBL 2024 Legal 2.0	EFW jump years treated
Georgia	GEO	Europe and Central Asia	50	77.5	2009–2014
Ghana	GHA	Sub-Saharan Africa	56.25	55	1985–1990
Greece	GRC	High income: OECD	83.125	87.5	1978–1983
Guatemala	GTM	Latin America and Caribbean	55	60	1992–1997
Haiti	HTI	Latin America and Caribbean	28.125	52.5	1977–1982
Honduras	HND	Latin America and Caribbean	52.5	65	2008–2013
India	IND	South Asia	35.625	60	1990–1995
Indonesia	IDN	East Asia and Pacific	43.125	60	1970–1975
Italy	ITA	High income: OECD	68.75	95	1970–1975
Kenya	KEN	Sub-Saharan Africa	60.625	70	2003–2008
Lao PDR	LAO	East Asia and Pacific	90	72.5	1989–1994
Latvia	LVA	High income: OECD	88.75	77.5	1997–2002
Lesotho	LSO	Sub-Saharan Africa	60	55	2001–2006
Lithuania	LTU	High income: OECD	91.875	85	1998–2003
Luxembourg	LUX	High income: OECD	90	85	1970–1975
Madagascar	MDG	Sub-Saharan Africa	54.375	50	2003–2008
Mali	MLI	Sub-Saharan Africa	37.5	47.5	2007–2012
Malta	MLT	Middle East and North Africa	87.5	77.5	1990–1995
Mauritius	MUS	Sub-Saharan Africa	67.5	87.5	1975–1980
Moldova	MDA	Europe and Central Asia	75	77.5	2013–2018
Morocco	MAR	Middle East and North Africa	66.25	60	1999–2004
Mozambique	MOZ	Sub-Saharan Africa	61.25	65	2000–2005
Namibia	NAM	Sub-Saharan Africa	74.375	67.5	1992–1997
Nepal	NPL	South Asia	47.5	62.5	1997–2002
Netherlands	NLD	High income: OECD	79.375	90	1985–1990
Nicaragua	NIC	Latin America and Caribbean	69.375	67.5	2010–2015
Nigeria	NGA	Sub-Saharan Africa	45	50	1970–1975
Paraguay	PRY	Latin America and Caribbean	46.875	80	1986–1991
Peru	PER	Latin America and Caribbean	58.75	85	1977–1982
Philippines	PHL	East Asia and Pacific	58.75	70	1982–1987

(Continues)

TABLE 10 | (Continued)

Economy	Economy code	Region	WBL 2024 EO 2.0 index	WBL 2024 Legal 2.0	EFW jump years treated
Poland	POL	High income: OECD	61.875	82.5	2012–2017
Portugal	PRT	High income: OECD	75	92.5	1972–1977
Rwanda	RWA	Sub-Saharan Africa	50.625	72.5	1993–1998
Saudi Arabia	SAU	Middle East and North Africa	85.625	50	2014–2019
Sierra Leone	SLE	Sub-Saharan Africa	52.5	72.5	2003–2008
Slovak Republic	SVK	High income: OECD	86.875	75	1999–2004
Spain	ESP	High income: OECD	81.25	90	1970–1975
Suriname	SUR	Latin America and Caribbean	62.5	65	1977–1982
Switzerland	CHE	High income: OECD	73.75	75	1977–1982
Taiwan, China	TWN	East Asia and Pacific	83.75	75	1995–2000
Togo	TGO	Sub-Saharan Africa	71.25	77.5	1972–1977
Trinidad and Tobago	TTO	Latin America and Caribbean	59.375	65	1980–1985
Türkiye	TUR	Europe and Central Asia	58.75	80	1997–2002
United Arab Emirates	ARE	Middle East and North Africa	81.875	62.5	2014–2019
Zambia	ZMB	Sub-Saharan Africa	78.75	72.5	1985–1990
Zimbabwe	ZWE	Sub-Saharan Africa	50.625	60	1972–1977

TABLE 11 | Impact of GDI Jump (≥ 0.10) on life expectancy.

	OLS (1)	Country FE (2)	Year FE (3)	Country + Year FE (4)
GDI Jump (≥ 0.10)	– 1.777*** (0.508)	0.805*** (0.261)	–2.742*** (0.471)	–0.252 (0.155)
Constant	66.136*** (0.114)			
Observations	8919	8919	8919	8919
R ²	0.001	0.001	0.004	0.0003

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors in parentheses.

in economic and social outcomes that one might expect. We interpret these findings through an institutional economics framework emphasizing how economic and cultural institutions often adjust to accommodate women's de facto participation before de jure legal recognition occurs. This adaptive capacity of economic institutions has been demonstrated in other contexts, such as responses to pandemics and natural disasters, where economically free societies show greater resilience and flexibility in adapting to shocks (Candela and Geloso 2021).

Further insights suggest that social welfare outcomes associated with economic growth emerge from the interplay of two broad sets of institutions. On the one hand, there are de jure legal institutions, defined by public protocols and governmental organizational structures; on the other, there are more informal institutions, encompassing socio-cultural norms, family dynamics, religious customs, and the broader social networks governing trade and exchange. At the margin, these forms of organization can substitute for one another, ultimately producing

TABLE 12 | Impact of GDI Jump (≥ 0.125) on life expectancy.

	OLS (1)	Country FE (2)	Year FE (3)	Country + Year FE (4)
GDI Jump (≥ 0.125)	− 3.468*** (0.657)	−0.493 (0.332)	− 3.616*** (0.610)	− 0.624*** (0.193)
Constant	65.938*** (0.135)			
Observations	6690	6690	6690	6690
R ²	0.004	0.0003	0.005	0.002

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors in parentheses.

TABLE 13 | Impact of GDI Jump (≥ 0.15) on life expectancy.

	OLS (1)	Country FE (2)	Year FE (3)	Country + Year FE (4)
GDI Jump (≥ 0.15)	− 3.478*** (0.692)	−0.220 (0.349)	− 3.793*** (0.642)	− 0.529*** (0.203)
Constant	65.924*** (0.135)			
Observations	6690	6690	6690	6690
R ²	0.004	0.0001	0.005	0.001

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors in parentheses.

TABLE 14 | Impact of GDI Jump (≥ 0.10) on maternal mortality.

	OLS (1)	Country FE (2)	Year FE (3)	Country + Year FE (4)
GDI Jump (≥ 0.10)	111.902*** (17.828)	13.315** (5.421)	112.528*** (17.801)	13.350*** (4.855)
Constant	170.923*** (4.528)			
Observations	3380	3380	3380	3380
R ²	0.012	0.002	0.012	0.002

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors in parentheses.

stable equilibria as communities negotiate the perceived costs and benefits of different arrangements. Within this context, GDI policy changes may represent either the authentic formalization of citizens' actual preferences or an exogenous disruption to a previously established institutional equilibrium. The latter would perhaps help explain temporal declines in economic and social outcomes, as new institutional patterns inspire reshaped behaviors and capital reallocation in the short run. This pattern was shown in O'Grady and Tagliapietra (2017), finding that state centralization of common-pool resources in the Italian Alps led to significant and persistent increases in infant mortality, with the most severe impacts in communities lacking prior

experience with formal institutions. In short, we find no strong evidence to support a the common view that enhanced formal state capacity in the form of de jure legal rights for women is responsible or deserving of credit for tangible improvements in female wellbeing or generalized economic and social progress.

In communities that have not historically embraced GDI norms, individuals who advocate for increased female participation in economic activities may look beyond de jure legal channels to validate women's productive capacities. Such efforts might manifest through informal or extra-legal strategies. Examples include but are not limited to: reconfiguring household production to

TABLE 15 | Impact of GDI Jump (≥ 0.125) on maternal mortality.

	OLS (1)	Country FE (2)	Year FE (3)	Country + Year FE (4)
GDI Jump (≥ 0.125)	203.800*** (24.978)	43.600*** (7.865)	199.617*** (24.989)	36.681*** (7.072)
Constant	174.942*** (5.125)			
Observations	2660	2660	2660	2660
R ²	0.024	0.012	0.024	0.011

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors in parentheses.

TABLE 16 | Impact of GDI Jump (≥ 0.15) on maternal mortality.

	OLS (1)	Country FE (2)	Year FE (3)	Country + Year FE (4)
GDI Jump (≥ 0.15)	192.274*** (25.347)	36.939*** (7.980)	189.393*** (25.362)	31.573*** (7.176)
Constant	175.644*** (5.131)			
Observations	2660	2660	2660	2660
R ²	0.021	0.008	0.021	0.008

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors in parentheses.

contribute more directly to GDP, making use of black markets shaped by migration and regulatory loopholes, or turning to grey market activities that subtly circumvent formal institutional constraints. These patterns highlight the complexity and adaptability of individual and group strategies across diverse and complex institutional environments. Meaningful shifts often unfold well before—or entirely outside—the realm of de jure legal recognition.

However, recent data from the Women, Business and the Law report presents an intriguing challenge to this interpretation. As shown in Table 10, there are notable gaps between cultural attitudes toward gender equality (WBL 2024 EO 2.0 index) and formal legal protections (Legal 2.0) across various countries. In some cases, such as Paraguay, Peru, and Mauritius, formal legal protections substantially exceed cultural attitudes, with differences of over 25 points on a 100-point scale.

This contemporary snapshot raises essential questions about our theoretical framework. If cultural and economic institutional progress truly precede legal changes, we expect cultural attitudes to align more closely with or exceed legal protections. The observed gaps might suggest that our understanding of the relationship between de jure and de facto economic participation requires refinement.

While legal frameworks may guarantee equal treatment, cultural norms and social expectations can significantly constrain women's economic participation (Papanek 1973; Bernhardt et al. 2018). In South Asia particularly, practices like purdah

restrict women's mobility and interaction outside the home (Kantor 2002), while deeply ingrained expectations of female domesticity create both explicit barriers and internalized constraints (Goldin 2021). Research shows that husbands' opposition to female employment often stems from concerns about social status and perceived violations of gender norms (Bursztyn et al. 2020), while women themselves may experience guilt about working outside the home (Dhanaraj and Mahambare 2022). These cultural constraints persist even across generations and levels of education (Dhar et al. 2022), suggesting that de jure equality alone is insufficient to ensure women's full economic participation.

However, this challenge comes with an important caveat: the Women, Business, and the Law data only provides a cross-sectional view from 2024. Without historical data on these cultural-legal gaps, we cannot determine whether these disparities were larger or smaller before the legal changes we examine in our study. If de jure institutional changes ultimately influence informal institutions, we would anticipate that any initial disparities between treatment and control groups would diminish, reflecting a gradual cultural convergence toward established norms. Likewise, if shifts in the Gender Disparity Index follow adaptive informal institutions, we would expect cultural incrementalism to align practices more closely with prevailing standards over time. Conversely, if the gaps were smaller or reversed, this would suggest a more complex relationship between de jure and de facto economic participation than our theory proposes.

TABLE 17 | Impact of GDI Jump (≥ 0.10) on GDP per capita.

	OLS (1)	Country FE (2)	Year FE (3)	Country + Year FE (4)
GDI Jump (≥ 0.10)	− 3876.965*** (980.783)	− 913.211* (532.064)	− 4588.812*** (965.899)	− 1721.189*** (483.063)
Constant	15,111.190*** (226.571)			
Observations	8601	8601	8601	8601
R ²	0.002	0.0003	0.003	0.002

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors in parentheses.

TABLE 18 | Impact of GDI Jump (≥ 0.125) on GDP per capita.

	OLS (1)	Country FE (2)	Year FE (3)	Country + Year FE (4)
GDI Jump (≥ 0.125)	− 6055.872*** (1238.557)	− 2504.162*** (657.627)	− 5710.277*** (1225.595)	− 2051.705*** (602.974)
Constant	14,664.850*** (262.419)			
Observations	6371	6371	6371	6371
R ²	0.004	0.002	0.003	0.002

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors in parentheses.

TABLE 19 | Impact of GDI Jump (≥ 0.15) on GDP per capita.

	OLS (1)	Country FE (2)	Year FE (3)	Country + Year FE (4)
GDI Jump (≥ 0.15)	− 6074.522*** (1306.136)	− 2432.891*** (692.479)	− 6009.067*** (1291.582)	− 2272.294*** (634.404)
Constant	14,637.090*** (261.821)			
Observations	6371	6371	6371	6371
R ²	0.003	0.002	0.003	0.002

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors in parentheses.

To provide a more robust defense of this theory, we utilize the Varieties of Democracy's variable property rights for women because it encompasses both formal legal rights and customary laws, religious codes, and social norms.³ Note that a change in GDI will necessarily be reflected in V-dem's property rights for women variable because it is an aggregate measure of the legal and social factors constraining property rights for women.

Figure 6 indicates a decline in gains for women over time; however, this may not be as clear when presented as year-by-year averages in Figure 7.

In these figures, we demonstrate that gains in a holistic view of female property rights were greatest before the legal system codified the changes occurring in society. If our theory were wrong, then the rate of change during the jump period would be the largest of the three presented. Instead, we find that the rate during the jump is somewhere in between the rate of change before and after the jump. Keep in mind that we are aware of the substantial gains made in the legal sphere during this period. After the jump period, the rate of progress for women is the lowest of the three periods. We can infer from this that social progress precedes legal recognition of female rights. It is also plausible that the codification of norm

TABLE 20 | Impact of GDI Jump (≥ 0.10) on urban sanitation.

	OLS (1)	Country FE (2)	Year FE (3)	Country + Year FE (4)
GDI Jump (≥ 0.10)	– 10.210*** (1.827)	0.276 (0.327)	– 10.207*** (1.830)	0.343 (0.272)
Constant	78.012*** (0.467)			
Observations	3182	3182	3182	3182
R ²	0.010	0.0002	0.010	0.001

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors in parentheses.

TABLE 21 | Impact of GDI Jump (≥ 0.125) on urban sanitation.

	OLS (1)	Country FE (2)	Year FE (3)	Country + Year FE (4)
GDI Jump (≥ 0.125)	– 15.676*** (2.464)	0.536 (0.431)	– 15.335*** (2.476)	1.129*** (0.358)
Constant	78.084*** (0.522)			
Observations	2493	2493	2493	2493
R ²	0.016	0.001	0.015	0.004

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors in parentheses.

TABLE 22 | Impact of GDI Jump (≥ 0.15) on urban sanitation.

	OLS (1)	Country FE (2)	Year FE (3)	Country + Year FE (4)
GDI Jump (≥ 0.15)	– 15.593*** (2.497)	0.448 (0.437)	– 15.341*** (2.510)	0.946*** (0.363)
Constant	78.061*** (0.522)			
Observations	2493	2493	2493	2493
R ²	0.015	0.0004	0.015	0.003

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors in parentheses.

changes calcifies social progress for women. Freezing gender rules could be beneficial if societies have reached full legal and social equality under the law. The countries receiving the jump treatment rarely approximate legal or social equality between genders.

Legal reforms that substantially alter traditional gender roles and economic rights may face resistance and implementation challenges that temporarily disrupt existing social support systems and economic arrangements before new institutional equilibria emerge. Similar to how macroeconomic shocks have heterogeneous effects across demographic groups (De

et al. 2021), institutional changes may disproportionately impact certain population segments during transition periods. This perspective suggests that policymakers should carefully consider the possibility of error and unintended consequences when managing transition periods and implementing major legal changes, perhaps by phasing in reforms gradually or providing additional support systems during periods of institutional flux. In general our findings when viewed in concert with established evidence would imply a more robust relationship between generalized economic freedom regardless of gender and social welfare outcomes over targeting policies specifically for gender advancement and equality.

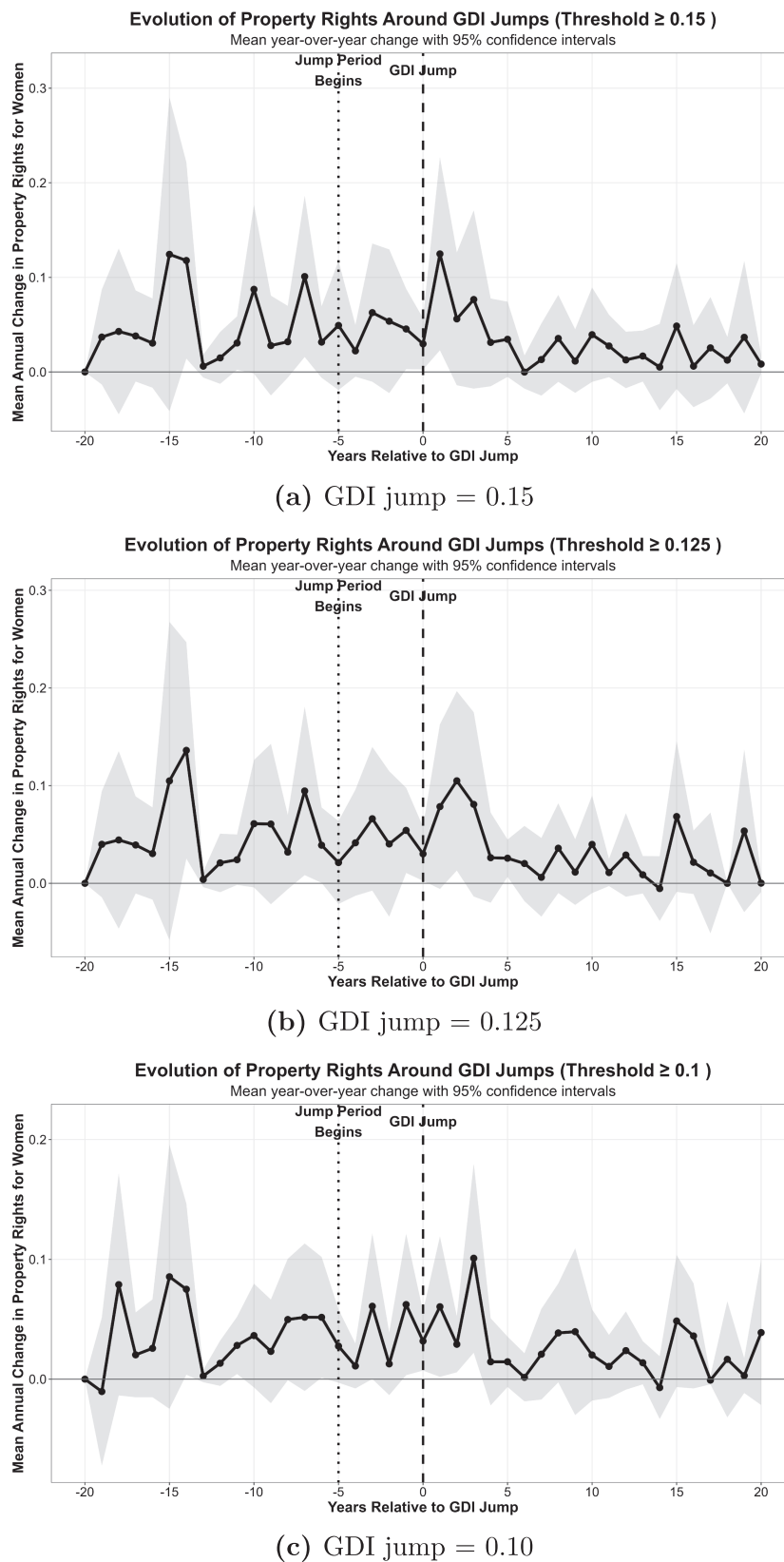


FIGURE 6 | Evolution of property-rights index under alternative GDI jump thresholds.

War can be a source of economic devastation and an inflection point for institutional change. Five countries in our jump table fall into this category. Indonesia's (IND) invasion and occupation of East Timor began in 1975 and resulted in a prolonged

and violent conflict during this period. The Nepalese Civil War (NPL) (1996-2006) was ongoing for most of this interval, involving intense fighting between government forces and Maoist rebels. Peru (PER) faced a major internal conflict with the Shining Path

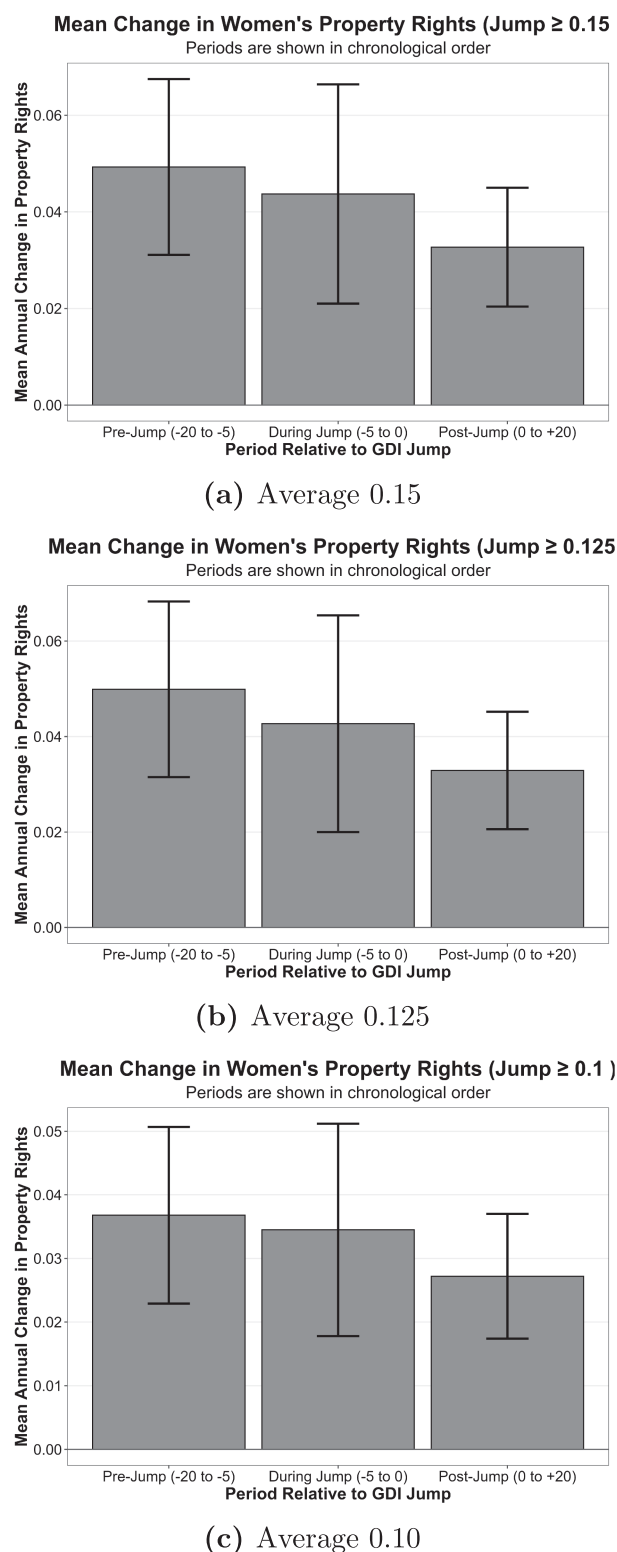


FIGURE 7 | Group-average property-rights index changes under alternative GDI jump thresholds.

(Sendero Luminoso) insurgency, which intensified in the early to mid-1980s. The Philippines (PHL) experienced ongoing armed conflicts involving both the Moro insurgency in the southern regions and communist rebel groups, constituting a significant internal conflict environment. Although the Rwandan (RWA) genocide ended in 1994, Rwanda remained heavily involved in regional

conflicts, most notably the Second Congo War (1998-2003), often considered one of the largest-scale and deadliest conflicts in recent African history. This explanation could be illustrative for explaining the treatment of a 0.10 jump in GDI, but none of these countries are in the primary treatment of 0.15. Therefore, we doubt that this resolves the dilemma.

Figure 8 shows the Economic Freedom of the World scores for jump countries, absent the GDI adjustment, over the time examined. The averages for a .15 jump and a .125 jump are the same year by year. We see a rise in overall economic freedom of the world scores in all counties included in our jump categories. Therefore, we cannot wave away our null results by saying there was a subsequent decline in other areas of economic freedom.

Legal changes toward greater gender equality likely resulted in a temporary disruption in labor markets, especially for women. It is plausible that we see women moving out of low-productivity sectors into newly opened and more productive careers. While we would expect this change to increase gross productive capacity, it is understandable if these results manifest after a longer time that allows for retooling and gaining human capital in the newly opened labor markets.

These uncertainties point to several important directions for future research: First, developing longitudinal measures of cultural attitudes toward gender equality would allow researchers to track how informal institutions evolve before and after legal reforms. This would provide a more rigorous test of our theoretical framework.

Second, detailed case studies of countries in Table 10 could illuminate the dynamic relationship between cultural attitudes, economic opportunities, and legal rights. Countries like Paraguay and Bolivia, which show significant gaps between cultural and legal measures, would be particularly interesting cases for understanding how formal and informal institutions interact. This observation could be an application of Boettke et al. (2015), particularly if advances in legal equality result from foreign pressures.

Third, research should examine whether the effectiveness of legal reforms varies based on the pre-existing alignment between cultural attitudes and formal institutions. This could help policymakers better understand when legal reforms will likely be most effective.

Fourth, formal property rights can paradoxically destroy wealth and social value in several key contexts, particularly in developing regions with established informal institutions. Leeson and Harris (2018) demonstrate this through their analysis of land privatization in Kenya, where formal property rights disrupted existing communal arrangements that had effectively managed resources. This aligns with Anderson and Hill's (Anderson and Hill 1983) earlier findings that the transaction costs of establishing and maintaining private property rights can outweigh their benefits, especially when existing informal systems already provide effective resource management. Murtazashvili and Murtazashvili (2021) further substantiate this through their research in Afghanistan, where they found that community-based land registration often proves more effective than

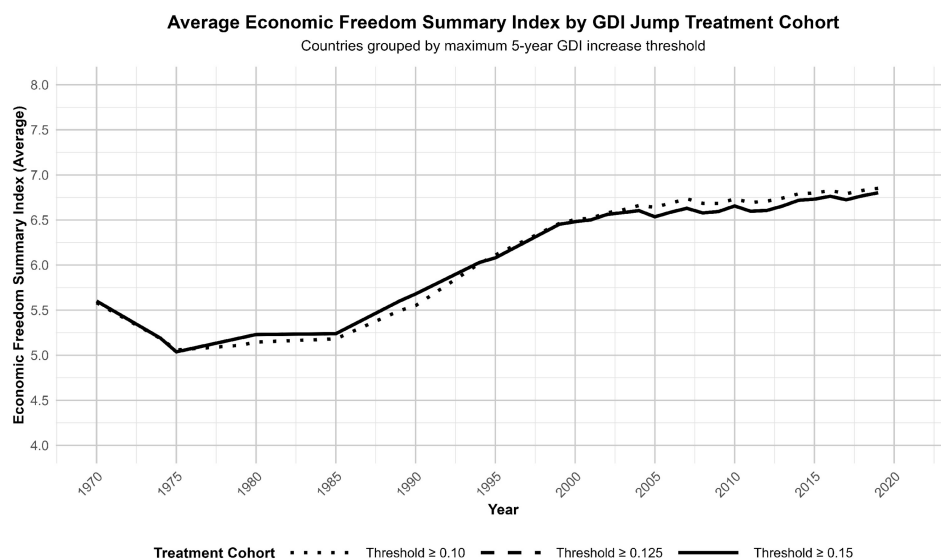


FIGURE 8 | EFW scores for jump countries.

state-mandated property rights, as these informal systems are better adapted to local conditions and social networks. In such cases, the imposition of formal property rights can undermine these well-functioning informal institutions, leading to increased conflict, higher enforcement costs, and ultimately, the destruction of social wealth. The papers cited above do not specifically discuss property rights concerning gender discrimination, but the mechanisms outlined in these papers may behave similarly in the context outlined in our paper. At a minimum, our empirical results raise the possibility of this, even if an empirical test of this hypothesis is impossible with our data.

While suggesting that formal legal changes may not be the primary driver of improved outcomes for women, our findings should not be interpreted as diminishing the importance of legal reform. Rather, they highlight the need to understand legal changes as part of a broader institutional ecosystem. Future research examining the complex interplay between formal and informal institutions will be crucial for developing more effective approaches to advancing gender equality.

Data Availability Statement

The data that support the findings of this study are available in EFW at <https://efotw.org/economic-freedom/dataset?geozone=world&year=2023&page=dataset&min-year=2&max-year=0&filter=0>. These data were derived from the following resources available in the public domain: - WBL, <https://wbl.worldbank.org/en/wbl%2Ddata> - PWT, <https://www.rug.nl/ggdc/productivity/pwt/?lang=en>.

Endnotes

¹Speech at the National Negro Business League, 1912. Walker (December 23, 1867 - May 25, 1919), an Indianapolis, IN-based entrepreneur, was the first female self-made millionaire in America.

²While negative rights in the GDI context involve freedoms from interference (such as not requiring male permission for work, travel, or financial transactions), positive rights require active provision of protections and opportunities (like guaranteed legal standing and business registration rights). The distinction is crucial because negative

rights generally require removing discriminatory barriers, while positive rights demand the creation of supportive legal frameworks and institutions. This distinction helps explain why some aspects of gender equality, like eliminating permission requirements, may be easier to implement than others that require building new institutional capacity, such as ensuring equal access to business licensing systems.

³Clarification: Private property includes the right to acquire, possess, inherit, and sell private property, including land. Limits on property rights may come from the state (which may legally limit rights or fail to enforce them); customary laws and practices; or religious or social norms. This question concerns the right to private property, not actual ownership of property. This question does not ask you to assess the relative rights of men and women. Thus, it is possible to assign the lowest possible score to a country even if men and women enjoy equal—and very minimal—property rights. Responses: 0: Virtually no women enjoy private property rights of any kind. 1: Some women enjoy some private property rights, but most have none. 2: Many women enjoy many private property rights, but a smaller proportion enjoys few or none. 3: More than half of women enjoy most private property rights, yet a smaller share of women have much more restricted rights. 4: Most women enjoy most private property rights but a small minority does not. 5: Virtually all women enjoy all, or almost all, property rights. Scale: Ordinal, converted to interval by the measurement model.

References

- Acemoglu, D., and J. A. Robinson, 2016. "Paths to Inclusive Political Institutions." In *Economic History of Warfare and State Formation. Studies in Economic History* edited by J. Eloranta, E. Golson, A. Markevich, N. Wolf, 3–50. Springer.
- Alvarez, S. P., V. Gelo, and M. Scheck. 2024. "Revisiting the Relationship Between Economic Freedom and Development to Account for Statistical Deception by Autocratic Regimes." *European Journal of Political Economy* 85: 102577.
- Anderson, T. L., and P. J. Hill. 1983. "Privatizing the Commons: An Improvement?" *Southern Economic Journal* 50, no. 2: 438–450.
- Bastos, J. P., and J. B. Pavlik. 2024. "Female Ownership of Firms and Regulation Experience." *Journal of Development Studies* 60, no. 9: 1412–1434.
- Becker, G. S. 1981. *A Treatise on the Family*. Harvard University Press Cambridge, Mass.

- Berggren, N., and C. Bjørnskov. 2024. "Does Capitalism Disfavor Women? Evidence From Life Satisfaction." *Southern Economic Journal* 91, no. 2: 436–464.
- Bernhardt, A., E. Field, R. Pande, N. Rigol, S. Schaner, and C. Troyer-Moore. 2018. "Male Social Status and Women's Work." *AEA Papers and Proceedings* 108: 363–367.
- Besley, T., and M. Ghatak. 2010. "Property Rights and Economic Development." In *Handbook of Development Economics*, 5, 4525–4595. Elsevier B. V.
- Boettke, P. J., C. J. Coyne, and P. T. Leeson. 2015. "Institutional Stickiness and the New Development Economics." In *Culture and Economic Action*, 123–146. Edward Elgar Publishing.
- Boockmann, B., and A. Dreher. 2003. "The Contribution of the IMF and the World Bank to Economic Freedom." *European Journal of Political Economy* 19, no. 3: 633–649.
- Burnette, J. 2024. "Are Some Piece Rates Better Than Others? Cross-Sectional Variation in Piece Rates at a US Cotton Factory." *Explorations in Economic History* 94: 101631.
- Burnette, J., and M. Stanfors. 2020. "Understanding the Gender Gap Further: The Case of Turn-of-the-Century Swedish Composers." *Journal of Economic History* 80, no. 1: 175–206.
- Bursztyn, L., A. L. González, and D. Yanagizawa-Drott. 2020. "Misperceived Social Norms: Women Working Outside the Home in Saudi Arabia." *American Economic Review* 110, no. 10: 2997–3029.
- Callais, J. T., and A. T. Young. 2023. "A Rising Tide That Lifts All Boats: An Analysis of Economic Freedom and Inequality Using Matching Methods." *Journal of Comparative Economics* 51, no. 3: 744–777.
- Callaway, B., and P. H. Sant'Anna. 2021. "Difference-in-Differences With Multiple Time Periods." *Journal of Econometrics* 225, no. 2: 200–230.
- Candela, R. A., and V. Geloso. 2021. "Economic Freedom, Pandemics, and Robust Political Economy." *Southern Economic Journal* 87, no. 4: 1250–1266.
- Chamlee-Wright, E. 1997. *The Cultural Foundations of Economic Development: Urban Female Entrepreneurship in Ghana*. Routledge.
- Chamlee-Wright, E. 2002. "Savings and Accumulation Strategies of Urban Market Women in Harare Zimbabwe." *Economic Development and Cultural Change* 50, no. 4: 979–1005.
- Coppedge, M., J. Gerring, C. H. Knutsen, et al. 2024). "V-dem codebook v14."
- Davis, L. S., and C. R. Williamson. 2019. "Does Individualism Promote Gender Equality?" *World Development* 123: 104627.
- Davis, L. S., and C. R. Williamson. 2022. "Individualism and Women's Economic Rights." *Journal of Economic Behavior & Organization* 198: 579–597.
- De, K., R. A. Compton, D. C. Giedeman, and G. A. Hoover. 2021. "Macroeconomic Shocks and Racial Labor Market Differences." *Southern Economic Journal* 88, no. 2: 680–704.
- Dhanaraj, S., and V. Mahambare. 2022. "Male Backlash and Female Guilt: Women's Employment and Intimate Partner Violence in Urban India." *Feminist Economics* 28, no. 1: 170–198.
- Dhar, D., T. Jain, and S. Jayachandran. 2022. "Reshaping Adolescents' Gender Attitudes: Evidence From a School-Based Experiment in India." *American Economic Review* 112, no. 3: 899–927.
- Doepke, M., M. Tertilt, and A. Voena. 2012. "The Economics and Politics of Women's Rights." *Annual Review of Economics* 4: 339–372.
- Duflo, E. 2012. "Women Empowerment and Economic Development." *Journal of Economic Literature* 50, no. 4: 1051–1079.
- Fafchamps, M. 2003. *Market Institutions in Sub-Saharan Africa: Theory and Evidence Volume 1 of MIT Press Books*. 1st ed. MIT Press.
- Faria, H. J., and H. M. Montesinos. 2009. "Does Economic Freedom Cause Prosperity? An Approach." *Public Choice* 141, no. 1: 103–127.
- Feenstra, R. C., R. Inklaar, and M. P. Timmer. 2015. "The Next Generation of the Penn World Table." *American Economic Review* 105, no. 10: 3150–3182.
- Fernández, R. 2014. "Women's Rights and Development." *Journal of Economic Growth* 19, no. 1: 37–80.
- Fike, R. 2016. "Gender Disparity in Legal Rights and Its Effect on Economic Freedom." In *2016 Economic Freedom Dataset, Published in Economic Freedom of the World: 2016*, 189. Annual Report.
- Goldin, C. 1994). "The U-Shaped Female Labor Force Function in Economic Development and Economic History." NBER Working Paper Series, pages 4707.
- Goldin, C. 1991. *The Role of World War II in the Rise of Women's Employment*, 741–756. American Economic Review.
- Goldin, C. 2006. "The Quiet Revolution That Transformed Women's Employment, Education, and Family." *American Economic Review* 96, no. 2: 1–21.
- Goldin, C. D. 2021. *Career & Family: Women's Century-Long Journey Toward Equity*. University Press.
- Grier, R. 2023). "The Causal Effect of Economic Freedom on Female Employment & Education."
- Gwartney, J. 2009. "Institutions, Economic Freedom, and Cross-Country Differences in Performance." *Southern Economic Journal* 75, no. 4: 937–956.
- Gwartney, J., and R. Lawson. 2003. "The Concept and Measurement of Economic Freedom." *European Journal of Political Economy* 19, no. 3: 405–430.
- Gwartney, J., R. Lawson, J. Hall, and R. Murphy. 2021). "Economic Freedom of the World - 2021 Annual Report."
- Hall, J. C., and R. A. Lawson. 2014. "Economic Freedom of the World: An Accounting of the Literature." *Contemporary Economic Policy* 32, no. 1: 1–19.
- Hasnas, J. 2024. *Common Law Liberalism: A New Theory of the Libertarian Society*. Oxford University Press.
- Horwitz, S. 2015. *Hayek's Modern Family: Classical Liberalism and the Evolution of Social Institutions*. Palgrave Macmillan.
- Jalota, S. and L. Ho. 2024). "What Works For Her? How Work-From-Home Jobs Affect Female Labor Force Participation in Urban India."
- Kantor, P. 2002. "Female Mobility in India: The Influence of Seclusion Norms on Economic Outcomes." *International Development Planning Review* 24, no. 2: 145–159.
- Kramer, C. W. 2024. "Economic Freedom and Legal Gender Disparity." *Contemporary Economic Policy* 43, no. 4: 573–588. <https://doi.org/10.1111/coep.12675>.
- Lamoreaux, N. R. 1986. "Banks, Kinship, and Economic Development: The New England Case." *Journal of Economic History* 46, no. 3: 647–667.
- Lawson, K. 2024. "Economic Freedom for Women and Fertility." *Contemporary Economic Policy* 43, no. 4: 638–651. <https://doi.org/10.1111/coep.12658>.
- Leeson, P. T. 2020. "Logic Is a Harsh Mistress: Welfare Economics for Economists." *Journal of Institutional Economics* 16, no. 2: 145–150.
- Leeson, P. T., and C. Harris. 2018. "Wealth-Destroying Private Property Rights." *World Development* 107: 1–9.

- Lemke, J. S. 2016. "Interjurisdictional Competition and the Married Women's Property Acts." *Public Choice* 166, no. 3: 291–313.
- Meagher, K. 2010). "Identity Economics: Social Networks and the Informal Economy in Nigeria, Volume 25 of African Issues." Boydell & Brewer, United Kingdom, ned - new edition.
- Mill, J. S. 1879. *On Liberty and the Subjection of Women*. Henry Holt and Co.
- Moehling, C. M. 1999. "State Child Labor Laws and the Decline of Child Labor." *Explorations in Economic History* 36, no. 1: 72–106.
- Moehling, C. M. 2002. "Broken Homes: The "Missing" Children of the 1910 Census." *Journal of Interdisciplinary History* 33, no. 2: 205–233.
- Moehling, C. M. 2004. "Family Structure, School Attendance, and Child Labor in the American South in 1900 and 1910." *Explorations in Economic History* 41, no. 1: 73–100.
- Murtazashvili, J. B., and I. Murtazashvili. 2021. "Land, the State, and War: Property Institutions and Political Order in Afghanistan." In *Cambridge Studies in Economics, Choice, and Society*. Cambridge University Press.
- Nardinelli, C. 1980. "Child Labor and the Factory Acts." *Journal of Economic History* 40, no. 4: 739–755.
- North, D. C. 1990. *Institutions, Institutional Change and Economic Performance. Political Economy of Institutions and Decisions*. 1st ed. Cambridge University Press.
- Nziku, D. M., and C. Henry. 2021. "Policies for Supporting Women Entrepreneurs in Developing Countries: The Case of Tanzania." *Journal of Entrepreneurship and Public Policy* 10, no. 1: 38–58.
- O'Grady, T., and C. Tagliapietra. 2017. "Biological Welfare and the Commons: A Natural Experiment in the Alps, 1765-1845." *Economics and Human Biology* 27: 137–153.
- Ostrom, E. 1994. *Governing the Commons: The Evolution of Institutions for Collective Action*. 1st ed. Cambridge University Press.
- Papanek, H. 1973. "Purdah: Separate Worlds and Symbolic Shelter." *Comparative Studies in Society and History* 15, no. 3: 289–325.
- Pavlik, J. B., and J. T. Callais. 2024. "Good for the Goose, Bad for the Gander? Corruption and Income Inequality." *Southern Economic Journal* 91, no. 3: 850–880. <https://doi.org/10.1002/soej.12733>.
- Rinne, U., and H. Sonnabend. 2021. "Female Workers, Male Managers: Gender, Leadership, and Risk Taking." *Southern Economic Journal* 88, no. 3: 906–930.
- Rode, M., and S. Coll. 2012. "Economic Freedom and Growth. Which Policies Matter the Most?" *Constitutional Political Economy* 23, no. 2: 95–133.
- Rode, M., and J. D. Gwartney. 2012. "Does Democratization Facilitate Economic Liberalization?" *European Journal of Political Economy* 28, no. 4: 607–619.
- Siegel, R. B. 1993. "The Modernization of Marital Status Law: Adjudicating Wives' Rights to Earnings, 1860-1930." *Geo. LJ* 82: 2127.
- Siegel, R. B.. 2018). *Home as Work: The First Woman's Rights Claims Concerning Wives Household Labor, 1850–1880*. In C. P. Gilman, 319–336. Routledge.
- Snowden, K. A. 1998. "Comments on Moehling, Siegler, and Wright." *Journal of Economic History* 58, no. 2: 548–552.
- Soto, H. D. 2003. *The Mystery of Capital: Why Capitalism Triumphs in the West and Fails Everywhere Else*. Basic Books.
- Stigler, G. J. 1992. "Law or Economics?" *Journal of Law and Economics* 35, no. 2: 455–468.
- Taveras, E. 2022. "The Effect of Spouses' Relative Education on Household Time Allocation." *Southern Economic Journal* 89, no. 3: 788–829.
- Williamson, C. R. 2009. "Informal Institutions Rule: Institutional Arrangements and Economic Performance." *Public Choice* 139, no. 3: 371–387.
- Wolcott, S., and G. Clark. 1999. "Why Nations Fail: Managerial Decisions and Performance in Indian Cotton Textiles, 1890-1938." *Journal of Economic History* 59, no. 2: 397–423.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. supplementary.pdf.